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Structuralist CGE Models

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Computable general equilibrium (CGE) models were first applied to developing economies in the early 1970s. Work has expanded rapidly since then, with two recent compendia (Devarajan, Lewis, and Robinson 1986; Decaluwe and Martens 1987) each listing fifty-odd models for two dozen countries around the world. It is clearly time to take stock of this effort, to see what has been learned and ask whether the CGE trail merits further pursuit. Robinson's (1989) survey paper is one such assessment. This book is another.

The emphasis here is on "structuralist" models, which account for only some of the CGEs. Their perspective is broader than that of other, more "neoclassical" constructs. Lustig (1988) nicely summarizes the ideas around which they are built: "Structuralist thought considers that structural characteristics (the repetition is appropriate) of the economy are fundamental to its behavior. Among the structural factors are the distribution of income and wealth, tenancy relationships on the land, the type and degree of specialization in foreign trade, the density of chains of production, the degree of concentration in markets, control of the means of production by distinct types of actors (the private sector, the state, or transnational capital), the functioning of financial intermediaries, and penetration of technical advance, as well as sociopolitical factors associated with the extent of organization of the working class and other influential sectors and classes, the geographical and sectoral distribution of the population, and its level of skills." These factors are at the foundations of the models in this book. Indeed, basing economic analysis on institutions and political economy is the hallmark of the structuralist approach. Neoclassical authors start from hypotheses of optimizing agents and full employment; their moves toward incorporating social classes and market power come as afterthoughts at best.

This chapter aims at showing how structuralist models work. It opens in section 1.1 with a discussion of how model builders strive to incorporate political economy into their machines. The organization of data for economywide models into social accounting matrixes is described in sections 1.2 and 1.3, first with a numerical example and then in terms of symbols. Possible directions of macroeconomic causality (or model “closures”) are outlined in a one-sector model without foreign trade in section 1.4. Additional causal links and the implications of opening the model to trade are taken up in sections 1.5 and 1.6, followed by a review of multi-sectoral data and closure issues in sections 1.7 and 1.8. Sections 1.9 and 1.10 show how the models can be extended to deal with flows of funds and finance, first in terms of data organization and then specification. Section 1.11 is about the treatment of nonproportionalities and substitution, and section 1.12 takes up dynamics. Section 1.13 reviews computational procedures for model solution. Finally, section 1.14 is a preview of the models presented in this book.

1.1 Political Economy

Structuralism is more a program of research and policy formation than a well-defined set of rules for putting a model in place. Any economy is a maze of structural detail—more than one could ever build into equations or use in policy design. However, general hypotheses about how the system functions are shared by structuralist authors. They include the following:

1. Economically powerful actors—“institutions” such as the state or corporations, “groups” such as landlords or rentiers, and “classes” such as organized labor—are not price-takers. They can influence price and/or quantity changes in certain markets. The seats of power differ from economy to economy, and change with local institutional arrangements and history. Identifying powerful actors and specifying the *inflexibilities* they can create is an indispensable first step in model design.
2. **Macroeconomic causality is influenced by** microeconomic detail, **but** on the whole it flows from investment, exports, and fiscal demand as predetermined variables (or demand “injections”) to income distribution, import, and output flows which generate savings supplies (or “leakages”). Changes in income and wealth distributions play an *essential* role in macro adjustment to historical processes such as capital accumulation and to exogenous shocks. There is no *particular* reason to assume that macro-

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economic equilibrium will involve full employment of either labor or installed capacity economywide.

3. The money supply is often endogenous or “passive,” adjusting to the level of activity and rate of inflation. In many sectors production responds to demand as just indicated, while the roots of inflation lie in unresolved distributional conflicts and propagation mechanisms such as the indexation of contracts.

4. Forms of financial intermediation that have developed over time strongly influence the macro system. Feedbacks between the real and financial sides of the economy can be mediated by large portfolio shifts (e.g., capital flight) as well as changes in asset rates of return. In developing economies the onset of “financial fragility” is an ever-present risk.

5. In **port due to the process of import-substituting industrialization**, imported intermediate and capital goods are required to support local production; foreign exchange to buy these imports can be a binding constraint.

6. Development is not a balanced or harmonious process. Technical change tied to new investment on the part of independently acting firms is an important contributor to growth. Private capital formation is as likely to be “crowded in” by public investment through complementarities as “crowded out” by rising interest rates charged by banks.

How are these views about the economy built into a model? **Structuralists’s** practice has five key features. First, they begin their analysis by singling out economically relevant sets of people and institutions and specifying how they fit into available data on income and wealth distributions. In Adelman and Robinson’s (1989) phrase, they create an “extended functional distribution” based on the institutional structure of the economy at hand. Each set of economic actors typically is related to a functional category of the income distribution (wages, corporate profits, receipts of unincorporated enterprise) or a production sector; it has different behavior patterns and partial controls over the system. In practice, a group, class, or institution may be labeled by specific income flows rather than membership in an identifiable cluster of firms or individuals, but that is a shortcut. The justification for taking it is the belief that actions both of legal persons such as the state or corporate sector and of real households reflect their positions in the economic system as revealed by their principal income source. They interact according to behavioral rules the modeler specifies, with consequent repercussions on macroeconomic equilibrium.

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portfolio choices, and power over price formation are assumed to differ. Typical entries in the distributional matrix include nonagricultural firms which gather profit income from markup pricing and undertake investment projects costing more than their own savings flows; rentiers who receive distributed **profits, interest, and other financial incomes, suffer capital losses** and gains on their (often considerable) assets, and save more than they invest; workers who get income from wages nominally fixed in the short run, don't save very much, and battle with firms about how wage increases will respond to unemployment and price inflation; agriculturalists whose saving rates are often high but whose income fluctuates sharply, following flexible price movements; and urban and rural "marginals" who pick up residual income flows, suffer from deprivation of basic needs, and save at low, often negative rates. The state in its fiscal, public investment, and central banking roles, the commercial banks, and foreigners also enter as partially independent and powerful actors. The degrees of freedom available to any actor depend on the institutions and history of the economy at hand; incorporating them in convincing fashion is part of the model-building art.

Starting from such notions of economic structure and institutions or class runs counter to neoclassical practice. Economists presently in the "mainstream" devote their time to "deriving" macro behavioral functions from stylized optimization exercises that individual persons, firms, and even the government are supposed to undertake. If clever, the analyst can fold the separate optimizations into a Pareto first-, second-, or nth-best equilibrium that can be represented as the extreme value of a social welfare function. A harmonious, welfare-maximizing economy (perhaps with a few "distortions" to be removed) represents the mainstream vision. Neoclassical modelers such as Scarf and Shoven (1983) or Shoven and Whalley (1984) reflect it in their computations by measuring welfare costs of distortions, or looking at effects of extra saving on growth. More adventurous neoclassicals may build in rent-seeking bureaucrats and other obstructions, but few go far in trying to bring class and economic power into their models through the rational agent back door.'

A **second distinctive feature of structuralist models is that they are not** set up in "real" terms (i.e., with only relative prices). Rather, they explicitly include prices and income flows in nominal or money terms, of the sort that actual firms, people, and ministers of finance face. Examples abound: Workers may be concerned about their "real" purchasing power (e.g., a wage index divided by a price index), but in practice they struggle or negotiate with management over a money wage. Rapidly rising money

wage rates in turn can trigger price inflations that affect everybody's economic decisions. Governments may be concerned with a "real" exchange rate (e.g., the ratio of indexes of prices of traded and nontraded goods), but they manipulate the nominal peso to yen conversion factor—**real or relative prices of traded goods may or may not respond. The finance minister worries about meeting payment obligations from a nominally fixed budget—if she underestimates inflation, money will have to be printed or the government's real spending will have to fall.**

Third, these examples suggest that different prices are under varying degrees of control by distinct groups in the economy—a focus on nominal magnitudes permits realistic treatments of financial markets and economic power. The models herein postulate that power plays a role in price formation. For example, econometric studies almost always show that firms (especially in manufacturing) mark up variable production costs by a fixed margin to fix prices. The markup rate is usually interpreted as the outcome of an oligopoly position on the output side and perhaps efficiency wage considerations for inputs. Other markets may have more flexibly varying prices, reflecting competitive behavior on the part of actors on both sides. The state may be able to intervene directly to control markups or public enterprise prices (e.g., the sale prices of gasoline and diesel fuel from refineries it controls) but finds controlling flex-prices costly in terms of holding stocks, importing, and "making" markets. Nominal price accounting provides the flexibility to include these and similar considerations in applied models.

Working with money income accounts also makes it easy to link the goods-producing and financial sectors of the economy through flows of funds. Each economic class or actor (e.g., wage earners, the government, or the rest of the world) has a "financial surplus" defined as the excess of its saving over its investment. The sum of nominal surpluses across the economy must be zero, though each one separately may be positive or negative. For example, rentiers and (perhaps) farmers usually save more than they invest, and industrial firms do the opposite. Among households, wage earners typically save far less than farmers or rentiers. The overall **public sector may save substantially or dissave, but after it pays for public investment, its financial surplus is usually negative.**

Shifts in surpluses depend on nominal price movements. Further price and quantity financial adjustments (at times involving asset and liability stocks that are the **cumulations** of flows) in response to changed positions can be built into CGE models. The feedbacks to commodity prices and quantities are often important. Effects of interest rate movements on in-

vestment demand are a classic example. Others involve portfolio movements and changes in wealth. For example, if rentiers hold assets abroad in the wake of capital flight, nominal currency devaluation will make them richer in local currency terms. A consequent reduction in their saving rate could have important macroeconomic implications, either for output changes or inflation.

Fourth, there is the question of how much economic rationality and price-mediated substitution to include in the specification. As a matter of convenience and common sense, the models here all incorporate some degree of consumer response to relative price shifts. If the staple food becomes more expensive, people will consume more or less of it, depending on income and substitution effects in complete systems of demand equations which are easy to build in. More important, dearer food may lead purchases of other goods to fall if its own demand is income inelastic. When there is a large agricultural sector, we will see in section 1.8 that the macroeconomic implications can be profound.

Elsewhere in the system, asset holders usually try to shift their portfolios to keep up with changing rates of return. Price-induced substitution among production inputs may be considered, if it seems practically relevant—econometric evidence rarely supports strong firm-level changes in labor demand as real wages vary, but there may be price-induced shifts among inputs such as copper and aluminum, citizens and guest-workers, or gas and oil. On the output side, the total volume of agricultural production is not very price responsive, but acreages devoted to specific crops are. Export volumes and investment are at times sensitive to prices and asset returns, depending on the economy at hand.

The moral from these examples is that price responsiveness may or may not be empirically important, or central, to the questions the model is supposed to address. If not, it can usefully be suppressed to focus attention on issues that are more relevant. A secondary observation is that one may or may not wish to characterize sensitivity to relative prices in an optimizing framework. Such machines are often convenient and neat but are not the end of the model builder's art. Consumers in an economy with several staple crops are likely to switch consumption to, say, rice from wheat flour if the price of wheat goes down, regardless of whether one believes they are doing careful optimizations or not. A model should include the demand switch in parsimonious fashion, not its utility-based textbook rationalization.

The fifth point about models is that their behavior depends crucially upon their description of causal linkages in the macroeconomic system. As discussed below, there are many ways in which the economy can adjust

when it is perturbed-output levels or the income distribution across classes may change, interest rates may vary, positions of wealth may expand or erode. A model builder has to select which of these adjustment mechanisms to build into his or her equations-the qualitative nature of the solutions will depend upon the choice. In the Jargon, a model's "closure" has to be chosen and justified on the basis of empirical and institutional analysis of the economy at hand. Setting closure is impossible unless class structures and economic power relationships have already been defined. Searching for sites of power and macro causal links is the key to the structuralist approach.

In summary, an emphasis on interactions between groups of households and institutions such as the state, banks, and firms characterizes the models in this book. Distributional shifts are the hinge for macro adjustment, mediated by output changes and movements of nominal prices against one another. How prices and production equilibrate is determined primarily by a model's causal structure, and only secondarily by substitution response. Estimating elasticities and elaborating substitution patterns in production or demand play second fiddle in model design to specifying its structures of causality and class.

1.2 The Data Base

CGE modelers make a fetish of accounting consistency. The numbers they feed into models must be consistent with the national income and input-output accounting their equations contain. The models are then used for comparative static exercises around the base data set or to make conditional projections for the future, all the while maintaining the accounting rules. It has become customary to impose tight bookkeeping from the start by arraying the data in the form of a social accounting matrix (SAM), of the sort proposed by Stone (1966) and elaborated by Pyatt and Round (1985) and many others. In this section we discuss SAM's for the nonfinancial part of the economy, postponing analysis of asset flows and stocks to sections 1.9 and 1.10.

Table 1.1 is a representative SAM for a one-sector economy-the numbers are fabricated, but for many developing economies their composition would be roughly right. They are put together subject to two main accounting considerations:

First, totals for corresponding rows and columns are always the same, a defining characteristic of a SAM. For example, total uses of output in row 1 (= 100) are the same as the first column's total production costs. Wage

Table 1.1
Illustrative numerical SAM for a one-sector economy

	Production costs 1	Uses of incomes				Export subsidy 7	Capital formation 8	Competitive imports 9	Total 10
		Firm 2	Wage 3	Rentier 4	Government 5				
1 Sales	30		25	4	12	23	2	-3	100
2 Wages	20				4		7		24
3 Profits	23								23
4 Indirect taxes	7								7
5 Import tariffs	2			1					3
6 Value added	52			1	4				57
7 Firm income	13								13
8 Wage income	20				4				24
9 Rentier income	10								10
10 Government income	9	3		2		5			19
11 Foreign income	18	5		1			5	3	32
12 Direct taxes		3		1					4
13 Subsidies			-2		4				0
14 Saving		5	1	3	-1	4			12
15 Totals	100	13	24	10	19	32	0	0	

income of 24 in row 8 is exhausted by the entries in column 3-workers's consumption of 23 (producers's price purchases of 25 in row 1 less a consumer subsidy of 2 in row 13) plus saving of 1 in row 14. The latter example shows that row and column numbers for corresponding income sources and uses do not coincide in table 1.1, for reasons of presentation to be explained below.

The second principle is that the matrix is set out in nominal or current-value terms. All entries in a row are valued in the same unit of account (typically an index based on producers's prices). Experience shows that the mapping from SAM to CGE is easier to grasp if the price used for "most" transactions along a row is the same. For example, row 1 gives the different uses of output, valued at producers's prices. It is assumed that the domestic price value of exports is 25, whereas foreign buyers benefit from an export subsidy of 2. The subsidy is in column 7, and the world price value of 23 is in column 6. They sum to exports in producers's prices.

Exceptions to the uniform price rule come up when there are well-entrenched price differentials across columns. For example, in the wage payment rows 2 and 8, production workers in column 1 and government employees in column 5 may receive different average remunerations. A purist would put the two sorts of wage payments in different rows to maintain single prices, but for such a simple case the extra space seems wasteful-one doesn't want to paper a wall with a SAM. It is easy enough to write down two separate wage rates in a model based on the social accounts.

The rows in table 1.1 are arranged in the following fashion: Sales by destination are in row 1. Components of value added at market prices are in rows 2 through 5, with totals in row 6. The payments entries are regrouped from production inputs (including indirect taxes) into income flows in rows 7 through 11. Besides receipts from productive activity, items such as transfers and direct taxes appear in these rows as well. Rows 7 through 11 correspond to the income recipient categories split out in the SAM-firms, wage recipients, rentiers who receive distributed profits from the firms, the government, and foreigners. Rows 12 and 13 summarize direct tax and subsidy flows, and row 14 gives savings from the different income categories.

In the columns, number 1 summarizes production costs, and the entries down numbers 2 through 6 show how the recipient groups dispose of their incomes. For example, column 6 gives foreigners' spending in border prices (foreign exchange prices converted to national terms by the exchange rate), for "our" exports in row 1, a transfer to the government of 5 in row 10, and

the current-account deficit (**or “foreign saving”**) of 4 in row 14. Foreigners’ inflows in row 11 are also **in** border prices. They include our imports of intermediates (**column 1**), **rentiers’ consumer** goods (column 4), capital goods (column 8), and “competitive” **items** (column 9) as well as **firms’s** transfers to abroad (**profit** remittances, etc.) of 5 **in** column 2.

The remaining columns include the export subsidy of 2 in number 7—note that it adds to **export** value **in** row 1 and is “**reflected**” through the negative entry to column 7, **row 13**, into the subsidy total of 4 in the government spending **column 5** (the **other** subsidy items is 2 for workers’ consumption in **column 2**). **Column 8 gives** investment spending (inventory changes are suppressed for **simplicity**), and column 9 shows that there are competitive imports of 3. Such commodities are assumed to substitute **easily** for domestic products. For **simplicity**, world and domestic prices for competitive imports **are assumed to be the same, although it would be easy** enough to include a price differential column like number 7. Of course, one would also have to specify to which income flow(s) the differential between world and domestic prices should be charged or accrue.

Other observations about table 1.1 include the following:

1. Errors and omissions are usually ironed away when a SAM is constructed. This procedure is at times inappropriate as, for example, when errors and omissions in the balance of payments are a proxy for rentier’s capital flight, offset domestically by borrowing and dissaving by the government or private firms.
2. Spending differentials across classes appear in the numbers. The share of wage income saved is lower than the shares from incomes of firms and rentiers. Rentiers consume imported goods (row 11, column 4) and also pay tariffs upon them (row 5, column 4). Workers’ spending in column 3 goes only for consumption of national goods in row 1, **with their costs reduced** in row 13 by a well-targeted subsidy.
3. Government **wage payments**, indirect **taxes**, and **tariffs** are included in GDP, following standard practice. The **wage** bill appears in the government-spending column 5, **adding** to the **labor** cost of production in column 1 **to give total wage income of 24** in row 2.
4. Investment in column 8 is split 7 to 5 into nationally produced and **imported** components, a plausible breakdown for a developing economy. The intermediate import share of gross output is 18 percent, **also** a reasonable number.
5. Saving flows from all sources in row 14 sum to total investment as a check on the **overall** accounting consistency of the SAM. In section 1.9 we

show how savings from the different income groups feed into flows of funds.

Lore about the fabrication of SAM's is presented in later chapters. Broadly speaking, the matrixes are based on national income and product accounts, interindustry data, and consumer budget information for the base year. Typically these sources are inconsistent, and judgment must be applied in modifying estimates that don't agree with one another to set up the consistent accounting of the SAM. The payoff comes from the fact that production and income numbers which tally with each other (even if they are "wrong") provide a powerful base for economic analysis. CGE models are designed to exploit the fruits of consistency to the greatest possible extent.

1.3 The SAM in Symbols

Table 1.2 restates the numbers of table 1.1 in algebraic notation, to pave the way for later economic analysis and also to show how many parameters in CGE models can be estimated (or "calibrated") as ratios of entries in the basic data source.

It is simplest to go through the table rowwise. Row 1 gives different uses of output with price P and volume X . Intermediate uses of domestically produced goods (or competitive imports) are proportional to production via the input-output coefficient a_1 . Workers's and rentiers's consumption volumes are, respectively, C_1^w and C_1^r . Government real consumption of domestic goods is G_1 , exports are E and competitive imports are M^c . Total real investment is I , and fractions β_1 and β_0 (summing to 1) are devoted to purchases of national and imported goods, respectively. The subscript 0 refers to an import.

A world price (and occasionally a foreign exchange stock or flow) is denoted by an asterisk. The world price of exports P_e^* comes from the relationship $P_e^* = P(1 - z_e)/e$, where z_e is the rate of subsidy. Rearrangement of this equation gives $P = eP_e^* + z_eP$, which explains the decomposition of the value of exports PE into world price and subsidy components in columns 6 and 7, respectively.

Going back to table 1.1, one would estimate the input-output coefficient a_1 by the ratios of entries in rows 1 and 15 of column 1: $a_1 = 30/100 = 0.3$. The export subsidy rate is figured relative to the producers's value of exports: $z_e = 2/25 = 0.08$. Finally, the share of national goods in the investment basket is $\beta_1 = 7/12 = 0.5833333$. Evidently $\beta_0 = 1 - \beta_1 =$

Table 1.2

Illustrative symbolic SAM for a one-sector economy

	Production ccsts 1	Firm's income use 2	Current spending by income type			Foreign income use 6	Export subsidy 7	Capital formation 8	Competitive imports 9	Total 10
			Wage 3	Rentier 4	Government 5					
1 Sales	$P_1 X$		PC_1^w	PC_1^r	PC_1^g	$eP_e^* E$	$z_e P^E$	$P\beta_1 I$	$-PM$	PX
2 Wages	wbX				$w_g L_g$					Y_w
3 Profits	$rP_1 K$									$rP_1 K$
4 Indirect taxes	$t_1 B X$									Ind
5 Import tariffs	$t_0 e P_0^* a_0 X$			$t_e e P_0^* C_0^r$						Tar
6 Value added	V			$t_1 e P_0^* C_0^r$	$w_g L_g$					GDP
7 Firm income	$(1 - d)rP_1 K$									Y_f
8 Wage income	wbX				$w_g L_g$					Y_w
9 Rentier income	$drP_1 K$									Y_r
10 Government income	T_z	T_f		T_r		U_{fg}				Y_g
11 Foreign income	$eP_0^* a_0 X$	U_{ff}		$eP_1^* C_0^r$				$eP_0^* \beta_0 I$	PM^c	Y_{for}
12 Direct taxes		$t_f^d Y_f$		$t_r^d Y_r$						Dir
13 Subsidies			$-z_e PC_1^w$		Z		$-z_e PE$			0
14 Saving		S_f	S_w	S_r	S_g	S_f		$P_1 I$		$P_1 I$
15 Total	PX	Y_f	Y_w	Y_r	Y_g	Y_{for}	0	$P_1 I$	0	

Definitions

$$B = Pa_1 + wb + (1 + t_0)eP_0^* a_0, \quad r = [P - (1 + t_1)B]X / (1 + t_1)P_1 K$$

$$P_1 = \beta_1 P + \beta_0 eP_1^*, \quad V = [wb + t_1 B + t_0 eP_0^* a_0]X + (1 + t_1)rP_1 K$$

$$T_z = [t_1 B + t_0 eP_0^* a_0]X, \quad T_f = t_f^d Y_f, \quad T_r = t_r^d Y_r + t_e eP_1^* C_0^r$$

0.4166667. These fractions are carried out to seven places because such precision would be wise in computational practice. In the numerical illustrations hereafter there is more rounding off for presentational ease.

Row 2 shows that wages are paid to production workers and government employees. **It is usually convenient to normalize base year prices** to unity: $P = w = \ell = 1$. The implication is that volume numbers are measured in units such as “rupees worth of output in prices of 1990.” Input coefficients such as the labor-output ratio can be computed as $b = \text{base-year employment}/X$. The government wage w_g could be given a value different from 1, to account for intersectoral wage differentials.

Skipping down to row 11, note in column 1 that intermediate imports are made proportional to output through a coefficient a_0 and (in row 5) pay a tariff rate t_0 . Such imports are usually assumed not to compete with domestic goods: They are “complementary” or “noncompetitive.” They have a world price P_0^* , which could be set equal to one in the base year. From table 1.1 the input coefficient is $a_0 = 18/100 = 0.18$, and the tariff rate is $t_0 = 2/18 = 0.1111$.

In the definitions at the bottom of table 1.2, B stands for variable cost per unit output—this concept is used to calculate a markup rate in section 1.4. The numbers in rows 1, 2, 5, and 11 of table 1.1 show that $B = (30 + 20 + 2 + 18)/100 = 0.7$. Indirect taxes in row 4 are figured on the basis of variable cost plus profits, so that the rate is $t_1 = 7/(70 + 23) = 0.0753$. By raising B , a higher tariff rate t_0 “cascades” into increased indirect taxes. Such effects occur often in developing country tax systems.

Row 3 of table 1.1 shows that total profits are 23. They give rise to a profit rate r in table 1.2, measured relative to the value of capital stock (at replacement prices) $P_i K$. The price index P_i is based on the investment proportions β_0 and β_1 . They sum to one, implying that $P_i = 1$ in the base year. If the capital stock is, say, 250, then $r = 23/250 = 0.092$.

We have now discussed the breakdown of value added in production, namely, V in row 6. To it must be added consumer tariffs and government wages to get GDP at market prices in the last column. The next step is to investigate income flows.

Foreigners' income in row 11 is made up of intermediate imports (column 1), firms' external remittances or transfers U_{ff} (column 2), rentiers' consumption imports C_0^r (column 4), investment imports with world price P_i^* (column 8), and competitive imports M^c (column 9). Competitive imports enter the accounting with a negative sign in row 1 since they offset demand or add to supply, and with a positive sign to contribute to foreign

exchange outflows in row 11. The consumer imports carry a tariff in row 5. Its rate is $t_c = 1/1 = 100$ percent.

Various taxes enter government income in row 10—they are defined at the bottom of the table. The government also gets a transfer from abroad, U_{fy} in row 10, column 6. Rentiers in row 9 receive a fraction d (“dividends”) of total profits. The rest goes to firms in row 7. From table 1.1, $d = 10/23 \approx 0.4348$.

Direct taxes are accounted for in row 12. The rates for firms and rentiers are $t_f^d = 3/13 = 0.2308$ and $t_r^d = 1/10 = 0.1$, respectively. Subsidies are cumulated in row 13, entering as a government expenditure item in column 5. The subsidy rate z_c on wage earners’s consumption is $z_c = 2/25 = 0.08$.

1.4 Macro Causality in the One-Sector Economy

With data and accounting rules exemplified, we are ready to discuss causal relationships in a one-sector CGE. As noted in section 1.1, structuralists often assume that macroeconomic causality runs broadly from demand injections to leakages under conditions of passive money supply.³ They treat the extended functional distribution and at least some sectors’s outputs as endogenous in their macro models, subject to rules such as markups and fix-price/flex-price distinctions (see section 1.8) regarding price adjustment. Neoclassicals, by contrast, typically presuppose full labor employment and capacity utilization under an “active” money supply that fixes the overall price level. They root their firms upon cost and input demand functions derived from clever optimization games.

Maintained hypotheses as diverse as these have notable effects upon the different models’s qualitative results. For example, greater investment demand in a structuralist model will make output rise and/or the income distribution shift to generate the complementary saving supply. With available saving determined from full employment output, distribution, and prices (or the forces of “productivity and thrift”) in a neoclassical world, an upward jump in investment is an impossible event. Such distinct outcomes arise throughout nonstructuralist and structuralist policy analysis, as we will see in discussing the contrasting effects under different causal schemes of exogenous changes in the income distribution as well as higher government spending, currency devaluation, and other policy moves.

The notation defined in table 1.2 serves as a basis to describe causal relationships in a one-sector CGE model or to define its “closure” (a term proposed by Sen 1963 and reinvented for the CGE field by Taylor and Lysy 1979). In formal algebra, prescribing closure boils down to stating

Table 1.3

Basic macro relationships for a one-sector model

Markup pricing	Neoclassical cost function	
$P = (1 + \tau)wb$	$P = f(w, rP)$	(1)
$L/X = b$	$L/X = f_w(w/P, r)$	(2)
$!A = X/K$	$u = X/K$	(3)
$r = [\tau/(1 + \tau)]u = \pi u$	$K/X = f_r(w/P, r)$	(4)
Investment-saving balance		
$g^s = sr$		(5)
$g^i = g^i(r, u)$		(6)
$\gamma + g^i - g^s = 0$		(7)
Definitions		
$\pi \approx rPK/PX = \tau/(1 + \tau), \quad w/P = (1 - \pi)/b, \quad y = PG/PK$		

which variables are endogenous or exogenous in an equation system largely based upon macroeconomic accounting identities, and figuring out how they influence one another. When one is setting up a model for any economy, the closure question becomes more interesting, transforming itself to one of empirically plausible signs of “effects” and, more important, a perception of what are the driving macroeconomic forces in the system. A sense of institutions and history necessarily enters into any serious discussion of macro causality. We can easily run through the algebra here; the models in this book give good examples of applied institutional perspective.

Table 1.3, based on table 1.2 but without its accounting complications, sets the stage. It contains two similar but not identical macro models—the neoclassical version to the right is itself more internally constrained or closed than the one with markup pricing. Properties of different closures can be illustrated by adding restrictions to the seven that appear, but first the table itself should be explained.

Consumption, investment, and government spending are the only components of demand, meaning that the material balance condition (comparable to row 1 in table 1.2) is

$$PC + PI + PG - PX = 0.$$

As written, this equation signifies that excess commodity demand vanishes. In what follows, we interpret it both as ex post accounting relationship that will be observed in a SAM and as a defining condition for macroeconomic

balance. Note for later reference that macro balance does not presuppose either full capacity use or full employment of labor (the two are not the same thing). Output X can in principle be determined by either effective demand or input supply.

For simplicity, we consolidate incomes of firms and rentiers. Then (ignoring intermediate inputs and indirect taxes) the income statement is

$$PX = wbX + rPK.$$

The saving rate from profit incomes rPK is s . We set workers's saving to zero, without much loss of generality in analyzing current output-expenditure interactions so long as the saving rate from wages is well less than that from profits.⁴ Total consumption becomes

$$PC = wbX + (1-s)rPK.$$

Combining the material balance, income, and consumption equations gives

$$PG + PI - srPK = 0.$$

This equality of private saving to investment plus government spending (or, more generally, the fiscal deficit⁵) is a familiar condition for macro balance. It is clearly not independent of the material balance condition-one is derived by accounting manipulation from the other.

Closure rules in a one-sector system are easier to illustrate if the foregoing equations are restated in a normalized or "intensive" form, with all nominal income and expenditure flows divided by the value of the capital stock PK .⁶ We will follow this practice until multisectoral models appear in section 1.7. We let $g^s = srPK/PK = s$ and $g^i = PI/PK$ stand, respectively, for the capital stock growth rate permitted by national saving and created by investment demand. Also let $\gamma = PC/PK$, that is, government outlay relative to the capital stock. Then the condition for macro equilibrium (or the investment-saving balance) is $\gamma + g^i - g^s = 0$, equation (7) of table 1.3. The growth rates possible from saving and investment appear in lines (5) and (6).

The arguments in the investment function are quite standard. Capacity use u represents the accelerator term that enters worldwide into econometric investment demand functions. Without going into fancy formulas for dynamic expectations (which don't make a lot of sense in applied models), the profit rate r in (6) can be interpreted as an index of expected future returns. Alternatively, it is a proxy for firms's available cash flow when investment is largely self-financed under credit rationing.

As the table headings indicate, we work with two pricing theories. Contrasting assumptions about pricing rules and the presence or absence of full employment or capacity use will provide the gist for the discussion that follows. On the left, the markup rule in row (1) simply states that output prices are based on variable costs (here, wages) with producers taking a margin at rate τ . In the more complete accounts of tables 1.1 and 1.2, the markup is defined by profit flows relative to variable costs: $\tau = rP_iK/BX = 23/70 = 0.3286$.

Markup pricing equations have the advantage of fitting the data, at least in nonprimary goods producing sectors where producers's margins are often stable. There is no dearth of theoretical explanations for both the level and stability of τ . They may follow from considerations of market power (Kalecki 1971; Sylos-Labini 1984), efficiency wages (Bowles and Boyer 1989) setting prices high enough to generate corporate savings sufficient to finance the bulk of planned capital formation (Duesenberry 1958; Eichner 1980), or ease in internal coordination in multiproduct, multiplant firms (Semmler 1984). The presence of a markup can thus be supported by optimizing models or—more plausibly—by empirical observation of enterprises and common sense. Its rate may change with the current macro situation, institutional evolution, and redistributive policy, as the models in the following chapters emphasize.

Following down the left-hand column, in line (2) the labor–output ratio is b , and the output–capital ratio u (which we treat as a measure of capacity utilization) is defined in row (3). With regard to (4), it is easy to see from (1) that the share of profits in output is $\pi = rPK/PX = \tau/(1 + \tau)$. Since the profit rate is the product of the profit share and the output/capital ratio u , (4) follows directly. This definition also implies that investment can be treated as a function of π (instead of r) and u .

The markup model (1) through (7) is a full macro system. It is of interest because markup pricing imposes few restrictions on supply. The markup rate τ and profit share π follow from institutions and policy, whereas the labor use coefficient b comes from technology and market custom. There is no mechanism via which τ (or other distributional parameters) and b interact. Such independence of distribution and resource allocation does not occur in the neoclassical model of the right-hand columns of table 1.3.

Equation (1) in the neoclassical model is a cost function. When production takes place under constant returns to scale (the standard hypothesis), the function f will be homogeneous; that is, it can be written in the form

$$1 = f\left[\frac{w}{P}, r\right].$$

This version is often called a “factor-price frontier” since *it sets up an* inverse relationship between the real wage w/P and the profit rate r . In the markup model such a conflict between real returns to labor and capital need not occur for macroeconomic reasons. The story goes as follows: The real wage is $w/P = 1/b(1 + \tau) = (1 - \pi)/b$, while from (4) r depends on τ and capacity utilization μ . A distributional shift toward labor would be represented by a lower τ (or π), directly increasing w/P . The overall saving rate would decline but (under expectations that the distributional shift is “permanent”) investment could fall as well if it responds more to a lower profit share than higher consumer demand. The net effect on output could go either way. With investment more sensitive to μ than π , the outcomes could include higher output, faster growth, and higher values of both r and w/P , something the factor-price frontier does not allow. These possibilities **underline the fact that the seemingly innocuous optimization exercises** underlying **neoclassical** production theory strongly constrain its results.

When some degree of input substitution is assumed, neoclassical equations (2) and (4) are based on a relationship known as Shepard’s lemma (cf. Varian 1984, 54–57), which states that factor input ratios are equal to the partial derivatives of the cost function, f_w and f_r . Equation (2) ties the ~~labor-output ratio to the real wage—it~~ w/P rises, L/X goes down. Labor use is linked to distribution. Greater employment is necessarily associated with a lower real wage—another strong neoclassical result.

From neoclassical equations (3) and (4), the output-capital ratio is positively related to the profit rate, as in the markup formula (4). The cost function (1) and the marginal productivity conditions (2) and (4) imply (and are jointly implied by) the existence of a constant returns production function, $X = g(L, K)$. When marginal productivity rules for input demands are postulated, a neoclassical model can be set up around either a production or cost function. For comparison with markup pricing, the latter is more convenient.

The causal structure of both versions rests on determination of price and input-output relationships from (1) through (4), and the level of demand from (4) through (7). How the two sets of equations interact is determined by additional restrictions, **or closure**. We discuss three standard **cases**—models characterized by output adjustment, forced saving, and determination of investment by saving supply—and in the following section consider additional adjustment mechanisms through which rigidities of the last version can be relaxed.

Output Adjustment

In the markup model assume that the nominal wage w is fixed by historical wage bargains and custom, the capital stock K by previous investment, and government spending G and (for the given K) the spending ratio γ by policy. Then with a fixed markup rate τ , the price level P and income distribution follow from (1). The demand equations (5) through (7) and the distributional identity (4) jointly determine r and u . Output X follows from (3) and employment L from (2). We have a macro system in which causality runs toward employment and output from the side of demand; output is implicitly assumed to lie below the full capacity level κK determined by the available capital stock. A classic formulation of the markup causal scheme is by Kalecki (1971).

The neoclassical story is similar, but more tightly constrained. Only γ , w , and K can be predetermined-distributional variables such as τ , the profit share, and the real wage are endogenous to the system. This complication is reflected in the fact that (1) must be combined with (4) through (7) to solve jointly for u , r , and w/P . Output and employment follow from (3) and (2) as before, whereas (1) gives the price level. Keynes's (1936) *General Theory* model of the real side of the economy is fundamentally the neoclassical equations (1) through (7) closed from the side of demand.

Comparative statics of the two versions can be illustrated with three experiments: changing government spending γ , the money wage w , and the real wage w/P .

In both versions a higher value of γ raises u and X through a multiplier process that converges so long as saving supply increases more strongly than investment demand as a function of u . In the neoclassical model, (i) **can be inserted into (4) to eliminate w/P . The upshot is a positive relationship between r and u .** But then from (i) interpreted as a factor-price frontier, w/P must fall as u goes up. An increased demand injection is associated with greater output and a lower real wage, with decreasing returns setting in as more labor is combined with the available capital stock. Since the nominal wage is fixed, P must rise to reduce w/P -the demand injection **drives up the price level as well as with output.**

Neither the real wage decrease nor the price increase occurs in the markup model since τ is set from the outside and diminishing returns do not set in so long as capacity exceeds output. A demand injection could cause real wage increases and price deflation if the markup rate or the labor-output ratio is an inverse-function of u , as some evidence suggests (Schor 1985). However, we do not pursue these possibilities here since they are not often included in CGE models.

A change in the money wage w has no real effects in the markup system. From (1), P just adjusts in proportion, and the real side is unchanged. The neoclassical equations can easily be seen to depend only on w/P ; therefore the same conclusion applies. For a given level of effective demand (i.e., y), **the real wage is an outcome of the neoclassical model.** Hence money wage changes will affect only the price level and not employment—this is basically Keynes’s argument about the uselessness of money wage reduction in chapter 19 of *The General Theory*.

Since the real wage is endogenous in the neoclassical system, it cannot be shifted exogenously in a thought experiment. In the markup model, if the rate τ is changed by macroeconomic shocks or policy (price controls, revised public enterprise charges, trade policy, etc.), the real wage will also move. As noted above, a real wage increase may make capacity utilization go either up or down. It will tell it there is a small differential between rentiers’s and workers’s saving shares and/or the responsiveness of investment demand to the profit share is strong. However, real wage increases will stimulate output if investment rises with the consumption surge that redistribution toward labor is likely to trigger. Through the same linkage, the growth rate may also go up.’

Dutt (1984) drew attention to this possibility for expansion, which can occur for additional reasons in an open economy model (section 1.6). It seems to apply in the developing world (Taylor 1988a).

Forced Saving

Forced saving occurs when μ and X cannot vary, for instance, because the economy is at full capacity (μ is at its “technical” maximum κ) or employment L is predetermined at a “full” level or otherwise. We work in the following discussion with a fixed L , but the analysis extends to a supply-side restriction on output by available capital or by a labor supply function such as $L = L(w/P, X, K)$. A surplus labor specification, for example, would make L highly elastic to the real wage, effectively setting w/P .

In the markup model, setting employment adds a restriction to (1) through (7). **In algebraic terms the system becomes overdetermined, and** some variable must become endogenous to meet the new constraint. If incipient excess demand with fixed output leads to upward pressure on prices, the obvious candidate for endogeneity is the markup rate τ , and thereby the price level P from (1). The causal structure is that (2) determines X , and (3) μ . The demand equations (5) through (7) give the profit rate r . The markup rate comes from (4) and P from (1). Through price movements relative to the fixed money wage (presupposing a greater saving propen-

sity from profits than wages), the income distribution adjusts endogenously to force the creation of enough saving to meet the new injection of demand.

In the neoclassical version, (1) and (4) through (7) determine u , r , and w/P as in the output adjustment closure. But then the model falls apart. Equations (2) and (3) give inconsistent values for X —the only recourse is that one equation be dropped. Two excisions generally appear in the literature. The first is to leave out the investment demand function, with implications to be taken up below. The other is to omit (2) and replace (4) by the identity $r = \pi u$. These moves are tantamount to abandoning marginal productivity input demand functions and returning to the markup system since $\pi = \tau/(1 + \tau)$.

This forced retreat underscores Kaldor's (1955) doubts about the usefulness of distribution theory based on marginal productivity. Alternatively, forced saving can be viewed as a macro adjustment scenario in which the neoclassical combination of output and price increases in response to higher demand degenerates into a pure price (and distribution) change when output is determined from the side of supply.

This last observation leads into comparative statics in the markup model. With u fixed, if the demand injection γ increases, then in (5) through (7) the profit rate is the free variable that goes up to provide the corresponding saving supply. From (4) and (1) the markup and price level rise and the real wage goes down. Reduced real spending (or "automatic lacking" in Dennis Robertson's 1933 phrase) on the part of wage earners caused by higher prices is the adjustment vehicle.

Since τ follows from effective demand, a money wage increase would just drive up the price level proportionally under forced saving. A worse outcome could befall an attempt at redistribution. Suppose in a simple case that investment is stable at the level $\bar{g}$ and that a tax $trPK$ is levied on profit income and transferred to wage earners. Saving-investment balance becomes $r = \bar{g}/s(1 - t)$. The profit rate rises along with the tax rate increase, and the real wage correspondingly falls.

Straightforward manipulation shows that the derivative of real post-transfer income per worker [which is $(w + trPK/L)/P$] with respect to t is $-t/(1 - t)$. If there is an existing tax-cum-transfer in place, increasing its rate worsens the position of wage earners. This is an example of the so-called paradox of thrift that accompanies forced saving. It bedevils any transfer attempt when the economy approaches full use of labor and/or available capital stock.

In closing, two observations should be added. First, although forced saving has been described here in terms of wages and profits, similar

redistributional effects can occur among any number of saving flows. Applied models will include saving from one or more income classes, the rest of the world, and the government. Movements in financial surpluses (saving less investment) as prices shift will differ across these classes and institutions, allowing forced saving to occur. The presence of flows fixed in nominal terms—transfers and state spending are common **examples**—makes redistribution induced by changing prices even more important.

Second, it bears note that the inverse effect of price changes on consumption demand resulting from forced saving was the adjustment mechanism used by the economists who Amadeo (1986) calls the “post-Wicksellian” generation, such as Schumpeter, Robertson, and the Stockholm school. It was the basis of Keynes’s (1930) *Treatise on Money*. After the publication of *The General Theory*, output adjustment came to the fore. When employment recovered in the industrial economies after World War II, macro theorists again wanted an inverse relationship between the price level and effective demand at hand. Instead of forced saving, a wealth effect was used to rationalize a negatively sloped schedule for aggregate consumer demand. Whether empirical relevance or ideological acceptability was the main reason for this choice would be an interesting inquiry, especially since we will see in section 1.5 that wealth effects provide a channel via which money wage cuts unambiguously make output rise.

Determination of Investment by Saving

Forced saving provides one means for accommodating a labor supply function into the macro system. As we have already noted, dropping the independent investment demand function (6) is another. This artifice is **widely used. For example, it is the heart of the neoclassical growth model of Solow (1956).**

Causal links are straightforward. In the markup model, K , L , w , and τ are predetermined. In table 1.3, P comes from (1), X from (2), u from (3), r from (4), and g^s from (5). In a savings-driven economy there is no room for (6), so capital stock growth g^i comes from (7). The solution of the neoclassical **version is similar, except that as usual there is no room for a predetermined** distributional index such as τ . With w fixed, equations (1) through (4) solve jointly for P , X , v , and u , and the story thereafter goes as in the markup model.

Output, income distribution, and saving all come from the supply side in the neoclassical version of this closure, which emphasizes productivity and thrift. A higher value of γ reduces g^i from (7)—government spending

“crowds out” investment demand. The real wage can’t be altered, and a money wage increase reflects itself solely in higher prices.

The markup model also demonstrates a pure price increase from a higher money wage and investment crowding-out by y . A real wage increase is represented by a lower τ . From (4) and (5) the profit rate and saving supply fall. Hence (7) shows that the growth rate must decline. Such outcomes are opposite those possible in the output adjustment closure discussed above. To reiterate, with potential results so different, empirical awareness and institutional understanding of the economy are required to judge how an apposite macro model should be closed.

1.5 Additional Adjustment Mechanisms

Since the publication of *The General Theory*, mainstream macroeconomics has devoted great efforts toward putting independent investment demand and labor supply functions into the same model-its theoretical pursuit of equilibrium full employment (ideally to be induced by real wage cuts) does not cease. In terms of table 1.3 this quest takes the form of adding endogenous variables or by postulating new adjustment mechanisms. The examples discussed in this section have all influenced neoclassical CGE work. They appear only rarely in models designed by structuralists, who are not wedded to the idea that full employment of resources is a part of macroeconomic equilibrium.

Adding Endogenous Variables to the System

In table 1.3 suppose that the investment demand function (6) is reinstated along with predetermined employment but that γ is endogenous. Then all equations can be retained, with γ coming from (7). The interpretation would be that the government regulates its spending to ensure full employment. Another possibility would be for the government to adjust some tax variable (e.g., the rate t in the tax-cum-transfer program discussed above) to the same end. The spending scenario is spelled out in Meade (1961), whereas Johansen (1960) implicitly used an endogenous income tax to permit both an independent investment function and full employment in the first practical CGE. Rattsø (1982) provides illuminating simulations around Johansen’s and other closures.

Postulating such government interventions perhaps make sense in models designed to explore distributionally “neutral” medium-run plans. In a shorter run, stock changes and/or trade flows such as competitive imports or

exports might vary to equilibrate demand with a predetermined domestic supply. Such balancing seems improbable at the aggregate level. Under a foreign exchange shock, for example, imports can only be reduced to zero, and export volume may be limited by world market conditions. However, **trade and/or inventory adjustments are more plausible for specific sectors** and are built into several of the models discussed in this book.

Loanable Funds

Changes in “the” interest rate (say, i) mediate an adjustment mechanism that has been suggested at least since Thornton’s and Ricardo’s day. If the saving rate s rises as a function of i while investment demand g^i declines, then all seven equations of table 1.3 comprise a well-behaved system. An exogenous **increase in γ** will make the left-hand side of (7) positive, and i will rise to restore equilibrium by cutting investment and inducing extra saving.

This story sounds sensible, but it is subject to at least two objections (both emphasized by Keynes). The first is structural, Rates of return to assets in principle are determined in markets for stocks, not by savings and investment flows. As we will see in sections 1.9 and 1.10, this problem can be addressed in practical models by cumulating flows of funds, so we leave it for the moment. Extending a CGE model to include financial linkages does not change the *directions* of effects of interest rate changes from those that happen with **loanable funds**.

The more serious question is about the *strength* of interest rate effects. There is not much evidence in the Third World that overall saving responds to interest rate movements (although portfolio compositions certainly do). With much investment controlled by the state and large enterprises with good access to finance, interest rate influences on aggregate demand will be limited to only part of gross capital formation (housing construction?) and perhaps consumer purchases of durable goods. Empirical questions obviously arise, but low elasticities can cripple adjustment based on interest rate movements. For that reason this mechanism does not usually figure in **applied CCE work**.

The Real Balance Effect

If output is predetermined and there is an independent investment function, consumption is the main demand component that must vary to permit macro equilibrium. If consumer demand is an inverse function of *the* price

level, the linkage is straightforward: Increased investment crowds out consumption by driving prices up. As noted in the last section, forced saving provides such a linkage but lost theoretical favor (apart from Kaldor and colleagues) fifty years ago. In contemporary macroeconomics its place is taken by the real balance effect.

The story is familiar, although, as we will see, it leads toward an **OVER-**determined model in a neoclassical specification. It rests upon an exogenous money supply, say, H . A price increase will reduce national wealth by eroding "real balances," H/P . Wealth holders might be expected to try to restore their position by saving more; hence the saving rate becomes $s = s(H/P)$. A higher P cuts consumption by increasing the saving rate. Crowding-out becomes feasible again.

In the markup model with variable output and employment, money wage cuts now have some bite. A lower w will reduce P via (1), increasing wealth and making consumption rise. Indeed, real balances were first given prominence by Pigou (1943) as a counterargument to Keynes's insistence that money wage reduction is of no use. If their impact on saving is strong enough, price-induced changes in real wealth make the "principle of effective demand" as enunciated by Keynes and Kalecki beside the point.

If L is predetermined while the markup varies, then the real balance effect complements forced saving since both mechanisms lower consumption demand. With an endogenously determined markup rate, more investment or state spending will crowd out real consumption by reducing both the real wage *and* real wealth.

In a neoclassical specification the argument when L is free to vary runs along the same lines as with markup pricing—money wage changes affect real balances by shifting the level of P through costs. However, with L and K **not exogenously or by supply functions, the model leaves no room for a** predetermined money wage. Equations (1) through (4) in table 1.3 solve for r , u , X , and w/P . Since the saving rate depends on H/P , the demand equations (5) through (7) set a value for P with the real balance effect in force. But then the money wage w has to be endogenous to permit **supply-side determination of w/P . There is no room left for historical or institutional elements in the wage bargain!**

These observations suggest that a short-run neoclassical specification incorporating predetermined employment along with the real balance effect makes no sense. In the long run the appropriate level of w can be made the equilibrium point of a wage adjustment process, for example, as in Friedman (1968), Fischer (1977), and Taylor (1980), but even these authors allow

determination of employment by aggregate demand until a vertical Phillips curve is reached. Their scenario is remote from the world of applied CGEs.

The problem with the real balance effect is that it picks out an absolute level of P as a function of the money supply H —it basically operates in the same way as the equation of exchange. In a full employment neoclassical economy all other nominal prices such as w must follow. Markup pricing allows two nominal magnitudes— w and H —to be fixed. Producers's margins and the overall price level take up the adjustment burden if full employment is imposed. If L can vary, it adjusts in place of τ . Ultimately a model builder must decide which of these mechanisms applies in the economy at hand. Using markup pricing instead of a marginal productivity black box forces one to think through one's prejudices about causality and class.

Similar considerations carry over to other liabilities and assets—valuations of stocks of wealth besides money are bound to change as prices adjust. As sensible financial analysts from Keynes (1931) and Fisher (1933) to Tobin (1980) have emphasized, all classes and *all* financial claims are affected by price movements in complex fashion. These changes and their consequent effects on economic behavior are only beginning to be addressed by CGEs. For what has been done, see sections 1.9 and 1.10.

The *inflation tax* is a dynamic version of the real balance effect. Inflation can only be described in models with more than one period, since it means growth of the price level over time. Its rate can be defined as $\hat{P} = [P(f) - P(f-1)]/P(f-1)$, where the arguments in f denote explicit time periods. An inflation tax makes saving a positive function of $\hat{P}$.

The rationale rests on real balances. When prices are rising, the reduction of one's real wealth over a period is given by $[P(f) - P(f-1)] \times [H/P(f-1)] = \hat{P}H$. It may be sensible to offset at least part of this loss by saving, so that the saving function becomes $g^s = sr + \alpha\hat{P}H$ with $\alpha \leq 1$. Faster inflation makes aggregate demand decline by acting as a tax on real balances.

This story is central to monetarist inflation theory. The full plot is set out elsewhere (e.g., Taylor 1988b), so we do not bother with it here. The effect at times is practically relevant, for example, in the consumption surges that have followed attempts to stabilize inflations by banning indexed contracts and imposing price controls (Taylor 1988a). For that reason it makes sense to incorporate an inflation tax into dynamic CGE models, where in structuralist variants inflation will stem from distributional conflict and propagation mechanisms, as discussed in sections 1.1 and 1.12.

Lance Taylor

Positive Effects of Inflation on Demand

Finally, inflation may activate demand. Suppose that i , the nominal loan rate of interest, is pegged or comes from the financial side of the model (section 1.10). Then faster inflation reduces the real rate $i = \dot{P}$, perhaps stimulating capital formation. If this possibility is deemed relevant, it is easy enough to build into an investment function. The same observation obviously applies to effects of inflation on competitive imports, demand for consumer durables, the efficiency of tax collection, or other variables.

1.6 Open Economy Extensions

The stylized data in tables 1.1 and 1.2 suggest that for short-run analyses the model just discussed should be extended to include exports and intermediate uses of imports and national goods. The demand-supply and cost equations become

$$PX = a_1 PX + PC + PI + PG + PE$$

and

$$PX = a, PX + wbX + rPK + eP_0^* a_0 X,$$

respectively. If we include saving from both rentiers (at rate s_r) and workers (rate s_w), the consumption function is

$$PC = (1 - s_r)rPK + (1 - s_w)wbX.$$

Putting these equations together gives an investment-saving balance of the form

$$PI = s_r rPK + s_w wbX + (eP_0^* a_0 X - PE) + (-PG).$$

National intermediates don't enter this accounting because they cancel out between the cost and demand decompositions. The saving counterpart of investment comes from several sources: rentiers ($s_r rPK$), wage earners ($s_w wbX$), the trade deficit ($eP_0^* a_0 X - PE$), and government dissaving ($-PG$). The last term would include taxes and other revenues as positive items if they were carried in the accounting.

These equations could easily be extended to incorporate consumer and capital goods imports, as well as emigrant workers's remittances and interest on foreign debt, which would have to enter into some domestic actors's income flows as positive and negative items, respectively. In numerical

Table 1.4
Macro relationships for a one-sector open economy model

Macro	relationships for a one-sector	open economy model
Markup pricing		Neoclassical cost function
$P = [1/(1 - \pi)](wb + eP_0^* a_0)$		$P = f(w, eP_0^*, rP)$ (1)
$L/X \approx L$		$L/X = f_w(w/P, eP_0^*/P, r)$ (2)
$M_0/X = a_0$		$M_0/X = f_e(w/P, eP_0^*/P, r)$ (3)
$u \approx X/K$		$u = X/K$ (4)
$r = AU$		$K/X = f_r(w/P, eP_0^*/P, r)$ (5)
Investment-saving and trade balances		
$\epsilon = e(eP_0^* P)$		(6)
$g^s = [\pi s_r + (1 - \pi)(\phi + (1 - \phi)s_w)]u - \epsilon - \gamma$		(7)
$g^i = g^i(\pi, u)$		(8)
$g^i - g^s \approx 0$		(9)
$\epsilon \approx (1 - \pi)\phi u = \epsilon t$		(10)
Definitions		
$A = rPK/PX$, $\phi = eP_0^* a_0 / (wb + eP_0^* a_0)$, $w/P = (1 - \pi)(1 - \phi)/b$		
$\epsilon = PE/PK$, $\gamma = PG/PK$, $r = T/PK$		
$s = KS_r + (1 - \pi)(\phi + (1 - \phi)s_w)$, $\Gamma = \eta\epsilon - (1 - \pi)s_w(1 - \pi)\phi u$		

models these extra foreign transactions would influence the role of the trade (or current-account) surplus in model responses as derived **below**—the formal modifications are left to the reader. Depending on the economy at hand, remittances, interest, and imports of capital goods would be the practically important foreign exchange flows.

In what follows, we will concentrate upon the comparative statics of changing two nominal magnitudes—the money wage w and exchange rate e . Even leaving out monetary considerations of the type discussed last section, these variables act through several channels and are subject to different institutional forces. For these reasons it is again helpful to work with a model set up in nominal terms. The equations appear in table 1.4, normalized by the value of the capital stock, PK . With minor loss of generality (rectified below), the markup and neoclassical price equations (1) net out intermediate uses of national goods as a component of cost.

The concept of output in table 1.4 is larger than value added. With national intermediates netted out, the cost equation in nonintensive form can be written as

$$PX \approx rPK + wbX + eP_0^* a_0 X,$$

which includes imported intermediates with their input coefficient a , and

world price P_0^* (for simplicity, tariffs are ignored). The import share of variable cost.

$$\phi = \frac{eP_0^*a_0}{wb - eP_0^*a_0},$$

turns out to be useful in analyzing the model. From this definition it readily follows that the import-capital ratio is $eP_0^*a_0X/PK = (1 - \pi)\phi u$, where π is the share of profits in output PX . The share of wages in the value of output is $(1 - \pi)(1 - \phi)$. Finally, the real exchange rate, e/P , is equal to $\phi(1 - \pi)/u$ so that responses of the model can be discussed equivalently in terms of ϕ or e/P (we stick with the former in the analysis that follows).

Since π enters the import-capital ratio directly, it is convenient to set up the markup function (1) using it as well. The version in table 1.4 follows from one based on a markup rate τ because $1 + \tau = 1/(1 - \pi)$.⁸ In the neoclassical version the cost function and its partial derivatives have the import cost eP_0^* as an argument.

An export function appears in equation (6), where $\varepsilon = E/K$. Exports are assumed to be sold in an imperfectly competitive world market. The world price of national goods (ignoring export taxes and subsidies) is P/e , while the price of foreign "similars" is P_e^* . Equation (6) shows that volume increases with devaluation (a higher e), and when local prices fall or foreign ones go up. This demand-based specification is broadly appropriate for industrial goods. It will not work well for primary commodities. A brief digression on how primary exports can be treated in CGE models follows.

Primary exports typically have a predetermined world price (call it P_e^* also), so that the f.o.b. local currency price is eP_e^* . This "border price" may or may not determine the internal market price P . Whether trade arbitrage and the "law of one price" are effective is an institutional question—in many primary exporting economies, institutions such as marketing boards rule arbitrage strictly out. If the two prices equalize, then the restriction $P = eP_e^*$ is added to the model. The markup rate (or, more generally, some national component of production cost) will have to be an endogenous variable from then on. If P and eP_e^* move independently, then their differential (multiplied by export volume) will represent a profit or loss for someone in the economy. It can be accounted for in the same way as the export subsidy in tables 1.1 and 1.2. Finally, depending on one's views about the market, primary export volumes may be fixed exogenously, or else determined endogenously as the excess of domestic supply over demand. The first option is a special case of (6) in table 1.4, whereas the other has already been noted as a convenient means to close a supply-determined model.

How currency devaluation works under these variant specifications is taken up below.

Returning to the discussion of table 1.4, equation (7) gives saving normalized by the value of capital. For later use, let s stand for the bracketed expression multiplying u , or saving per Unit Of capacity use:

$$s = \pi s_r + (1 - \pi)[\phi + (1 - \phi)s_w].$$

Equation (8) is the investment function, and (9) defines macroeconomic balance. If T is the trade surplus in foreign currency, and $t = T/PK$, then the trade balance is (10). Depending on closure, T may or may not be endogenous. Indeed, to avoid tedious taxonomy, we consider only two closures here. The first has u determined by demand and T endogenous. The second fixes T exogenously (a binding trade constraint) and asks how the rest of the system adjusts.⁹

Output Adjustment

Either higher government spending γ or an upward shift in the export demand function ε will increase capacity utilization in the markup model—in this regard the open economy version differs little from table 1.3. More interesting are the effects of movements in the key nominal prices, the exchange rate and the wage. Since their effects through π and u on investment demand are straightforward, in the following discussion g^i is held constant.

Nominal devaluation influences the system in several ways. From (1), it pushes up the price level with elasticity ϕ , shifting income shares from workers toward higher saving rentiers and the rest of the world. Since ϕ itself is an increasing function of θ (the elasticity is $1 - \phi$), (7) shows that devaluation raises saving due to intermediate imports while cutting back on wage earners's saving by a lesser amount (since $s_w < 1$). Aggregate demand falls—an example of forced saving at work. Finally, exports respond to a higher ℓ (and a lower P) with elasticity η .

Putting these effects together in equations (6) through (9) shows that devaluation increases u when $\Gamma - \eta\varepsilon \sim (1 - s_w)(1 - \pi)\phi u$ is positive. A bit of manipulation shows that this condition is equivalent to

$$\eta > \frac{(1 - s_w)\varepsilon P_0^* a_0 X}{PE}.$$

Output expansion is likely with high values of the export elasticity η , the workers's saving share s_w , and the ratio of exports to imports. If the

world demand elasticity and workers's saving are low or the trade deficit is large, output contraction after devaluation is likely to result. Such a response shows up econometrically in many developing countries, as well as in structuralist CGEs.

The money wage enters the present model as a mirror image of the exchange rate, affecting all variables in (6) through (9) with the same magnitude but opposite sign. A higher w makes output increase when devaluation is contractionary, when $\Gamma < 0$ or the inequality above is violated. This result suggests that in the Third World, last section's arguments in support of money wage cuts to raise employment should be taken with a grain of salt. Output expansion in response to higher wages shows up in applied CGE work for the reasons just discussed (not to mention the investment responses mentioned in section 1.4), although we will see shortly that strict opposite sign symmetry of e and w is broken in more complicated specifications.

If prices are formed according to a neoclassical cost function, the analysis becomes more complicated. It can be illustrated with an example. Suppose that national and imported intermediates trade off for one another with a constant elasticity of substitution σ . As explained in section 1.1.1, this basically means that the intermediate input demand functions can be written in the form $a_1 = \bar{a}_1(P/Q)^{-\sigma}$ and $a_0 = \bar{a}_0(eP_0^*/Q)^{-\sigma}$, where the "bars" indicate constant terms and Q is a CES aggregate cost function for intermediates: $Q = Q(P, eP_0^*)$. An increase in e will raise Q with an elasticity equal to the cost share of imports (say, ξ) in total intermediates. Elasticities of a_1 and a_0 with respect to e are $\sigma\xi$ and $-\sigma(1 - \xi)$, respectively.

Now consider a devaluation. It increases the cost ratio ϕ directly but also makes it decline by reducing a_0 as national goods become cheaper relative to imports. The increase in potential saving from a higher ϕ is less than in the case with no substitution, so output contraction becomes less likely. Formally, the condition for expansion becomes

$$\eta > [1 - \sigma(1 - \xi)](1 - s_w)eP_0^*a_0X/PE,$$

which is more easily satisfied the higher is σ . If σ takes an empirically plausible value (well less than 0.5), the inequality may still not hold.

The story in this neoclassical model about wage increases goes the same direction as in the markup specification; however, the opposite sign but equal magnitude effects of w and e don't occur for substitution only between intermediate inputs. Substitution between labor and imported intermediates would make symmetry come back. It is a matter of judgment which version is more realistic. Regardless of which applies, dedicated

econometricians have a hard time coming up with high values for the relevant elasticities of substitution in most economies of the Third World.

A Binding Trade Constraint

Frequently in developing countries, it seems as if trade deficits beyond a certain level cannot be financed. At the same time, political forces make aggregate demand (and therefore intermediate imports) impossible to cut back. In such a case of a binding trade constraint, capacity utilization will be determined from equation (10) of table 1.4, with a fixed value (negative or positive, depending on the external debt situation) of the dollar trade surplus T . Formally, we have $u = (e - et)/(1 - \pi)\phi$, where $f = T/PK$.

Another restriction has been added to the model. As usual we have to come up with a corresponding endogenous variable. One possibility is the nominal exchange rate itself, although if there is a nonconvertible currency (the usual case in the Third World), it is more naturally treated as being a policy tool. The obvious alternative is the profit share π , especially when we observe that adverse external shocks in developing countries are usually followed by output stagnation and inflation. Forced saving (perhaps supplemented by real balance effects or inflation taxes) is the adjustment vehicle. Any foreign transfer or capital inflow will reduce t , permitting u to increase and cutting back on prices. The relevance of this scenario is corroborated by the anti-inflationary effects of foreign resource transfers that have been observed in externally strangled developing countries in recent years as well as the inflations that spring up when inflows are cut off (Taylor 1988a).

What will be the effects of devaluation in these circumstances? It is simplest to tell the story in terms of the trade constraint, which can be rewritten as

$$X = \frac{(P/e)E(eP^*/P) - T}{a_0 P_0^*}.$$

The only terms that depend on the exchange rate here are the price level P , export volume E , and the input coefficient a_0 . Devaluation will permit X to increase if

$$\frac{PE}{ea_0 P_0^* X} (\eta - 1)(1 - \phi) + \sigma(1 - \xi) > 0.$$

Broadly speaking, if σ is near zero (intermediate input substitution is weak), then the demand elasticity η will have to be not far below one to

permit output to expand. The reason is simple. Devaluation reduces P/e , the foreign price of exports. It must generate an elastic volume response to offset this terms of trade loss. Such reasoning underlies trade theory's familiar Marshall-Lerner condition (which can be derived from the inequality above).

The effect of e on the profit share π is more complex. It goes through equation (7), in which devaluation affects exports E , capacity utilization u , and the import cost share ϕ all at once. Leaving aside intermediate input substitution, one can show analytically from (7) through (10) as a simultaneous system that π will rise with devaluation if

$$\Gamma(1 - \pi)\phi - s(\eta - 1)E > 0.$$

This expression is not terribly illuminating. However, if we focus on the export elasticity η as a key parameter, it can be restated as

$$\eta < \frac{s - (1 - \pi)\phi[(1 - s_w)(eP_e^*u_0X/PE)]}{s - (1 - \pi)\phi}$$

The bracketed term multiplying $(1 - \pi)\phi$ in the numerator will exceed one if, is small and there is a substantial trade deficit. Therefore the upper bound on η is less than one. A value below the bound means that devaluation cuts back on u , increasing excess demand and exacerbating forced saving. A larger elasticity means that devaluation generates enough foreign exchange to allow forced saving to abate. Only detailed computation would reveal which way the influences run.

Finally, two words should be added about the effects of exchange rate changes when primary product exports are included in the model as discussed above. First, if the relevant sectors use imported intermediates, a change in e will affect their cost structures, with implications similar to those just discussed.

Second, the domestic price level may be related to the foreign one via the law of one price $P = eP_e^*$ or some more complex formulation. (For example, P might be made more or less elastic to eP_e^* depending on the cash position of an export marketing board.) If, further, supply in the export sector responds to the price/cost ratio as discussed in section 1.11 and export volume is the variable that adjusts to bring sectoral demand-supply balance, then devaluation will stimulate exports by raising output, while perhaps having a negative effect by driving up costs. The outcome will depend on details of the specification, subject to as many uncertainties as in the "industrial" export version. CGE models are of use not because

they give the “right” quantitative assessments about policy responses but because they help one trace the directions in which they are likely to run.

1.7 Models with Many Sectors

The last exercises suggest that algebra is reaching its limits in deriving comparative static results. In this and the following section, we shift focus to discuss in more qualitative fashion the data and closure questions that arise when a CGE model contains more than one sector. Sections 1.9 and 1.10 take up financial accounts and models based upon them.

Table 1.5 gives two sectors’s worth of SAM, suppressing government transactions to save space. The first sector is called “agriculture” and the other “industry” for reasons that will become clear in section 1.8. All agricultural value added is lumped into that sector’s income Y_a , and industrial incomes are split into wages and profits as before. The underlying assumption is that income classes by sector have disparate saving and consumption patterns, as discussed in section 1.1. Columns 3 through 5 are dedicated to these differences.

Two sets of parameters in the intersectoral accounts stand out in the table, and two more are implicit. The most visible are the input-output coefficients a_{ij} , where i runs from 0 to 2 (or N , in an N -sector model), and j from 1 to 2 (or N). As before, the 0 subscript denotes a foreign input, of intermediate goods in this case. These coefficients can be estimated as ratios from the base year flow accounts of a SAM, as discussed in section 1.3. Indeed, SAM’s are usually put together for years in which an input-output table has been constructed (or seriously updated from the past), so the a_{ij} represent an essential part of a model’s data base.

The other explicit parameters are the capital coefficients β_{ij} . The investment flows I_1 and I_2 in table 1.5 stand for real gross capital formation “by destination,” namely, in the sectors in which the investment takes place. However, sectoral demand-supply accounting is done “by origin” or “source,” namely, in terms of the sectors producing the relevant goods. The β_{ij} coefficients transform investment by destination into investment by source.

Information about capital coefficients is usually far more scanty than about their input-output cousins. They typically are pieced together on an ad hoc basis from industrial and agricultural cost surveys and similar sources. Lacking such data, a common artifice is to make all columns of the β -matrix the same. Each column must sum to one for accounting consistency, and proportions between the entries could come from the values of

Table 1.5
Illustrative symbolic SAM for a two-sector economy

	Production costs		Uses of incomes				Gross capital formation	Totals
	Agriculture	Industry	Agriculture	Wage	Rentier	Foreign		
	1	2	3	4	5	6		
Sales								
1. Agriculture	$P_1 a_{11} X_1$	$P_1 a_{12} X_2$	$P_1 C_1^a$	$E_1 C_1^w$	$P_1 C_1^r$	$P_1 E_1$	$P_1 (\beta_{11} I_1 + \beta_{12} I_2)$	$P_1 X_1$
2. Industry	$P_2 a_{21} X_1$	$P_2 a_{22} X_2$	$P_2 C_2^a$	$E_2 C_2^w$	$P_2 C_2^r$	$P_2 E_2$	$P_2 (\beta_{21} I_1 + \beta_{22} I_2)$	$P_2 X_2$
Incomes								
3. Agriculture	V_1							Y_a
4. Wage		$w_2 b_2 X_2$						Y_w
5. Rentier		$r_2 P_1^r K_2$						Y_r
6. Foreign	$e P_0^{*a} a_{01} X_1$	$e P_0^{*a} a_{02} X_2$					$e P_1^*(\beta_{01} I_1 + \beta_{02} I_2)$	Y_{for}
Saving								
7. Saving			S_a	S_w	S_r	S_{for}		I_{tot}
Totals								
8. Total	$P_1 X_1$	$P_2 X_2$	Y_a	Y_w	Y_r	Y_{fr}	I_{tot}	
Definitions								
$P_j^i = P_1 \beta_{1j} + P_2 \beta_{2j} + e E^{*a} \beta_{0j}, \quad j = 1, 2$								
$I_{tot} = P_1^r I_1 + P_2^r I_2$								

investment flows by origin (e.g., P_{ij}^i , $i = 0, 1, \dots, N$) from the input-output table. This assumption of identical columns could be relaxed to capture broad **sectoral** differences; for example, if one has independent information on the makeup of the capital stock in agriculture, then the nonagricultural capital goods composition can be denved from **economywide** totals as a residual. In any case, only a few industries produce capital goods (e.g., machinery, construction, producers's services, and imports), so most rows of the **β -matrix** in a model with more than a few sectors will be zero.

Capital coefficients (ideally set up with a time dimension to reflect investment gestation lags) are essential parameters if one wants to cumulate investments by destination into capital stocks over time in a dynamic model as described in section 1.12. On the other hand, they can be dispensed with in comparative statics, if one sticks with investments by origin as the relevant components of final demand. The other main use of the β_{ij} in the models of this book is to set up capital goods price indexes P_j^i (the superscript stands for "investment" and the subscript for sectors) which are used to figure costs of investments by destination and value capital stocks for use in calculating rates of profit. Absent a full matrix, using the same set of coefficients for all sectors can suffice for these purposes.

The first set of implicit parameters in table 1.5 enters into consumer demand functions. To see what is involved, note that column 3 can be written in the form $P_1 C_1^a + P_2 C_2^a = Y_a$, $S_a = D_a$, where D_a stands for total consumption spending from agricultural sector incomes. As the models in previous sections illustrate, variables such as D_a are functions of income flows (e.g., Y_a) and saving and tax parameters. Somehow in the model specification, a value sum of consumer purchases like $P_1 C_1^a + P_2 C_2^a$ must be forced into equality with the corresponding spending flow.

The simplest means to enforce consumption adding-up (or "Engelaggregation") is complete system of demand equations based upon the neoclassical pretense that consumers maximize utility subject to a budget constraint—the constraint is what forces outlays by sector to sum to total spending. A clever choice of utility function gives convenient equations. Since it is easy to use and interpret, the models in this book mostly adopt a **specific formulation called the "linear expenditure system" (LES)**. How the LES is **parameterized** is discussed in section 1.11.

Finally, similar adding-up restrictions are implicit in columns 1 and 2 of table 1.5. Total costs must sum to $P_i X_i$, and the components of value added to the correct total. The first requirement is satisfied if the numerical model is based upon ratio estimates of input-output coefficients, tax and tariff rates, and so on, from the base-year SAM. With markup pricing, computa-

tion of the markup rate from the SAM takes care of adding-up for value added. The assumption of constant returns to scale does the same for a neoclassical cost function. The function's scaling parameters will be determined by base-year data, as described in section 1.11.

1.8 Closure Issues with Many Sectors

Multisectoral models repeat in many dimensions the questions about how economic classes are defined and interact that we have already addressed. They can be phrased in terms of the circular flow of income from production to demand and back to the level of output. From this perspective each sector in a CGE model requires a four-part specification:

1. Its value added (plus imported intermediates) must be broken down into income flows that fit the issues the model is trying to address. Also needed are functional relationships via which these flows are determined from the output level and input prices, subject to the column identity in the social accounts stating that the sector's value of output must just be exhausted by costs.
2. Rules for mapping input costs into price formation must be specified. Possible formulations include a markup specification or some sort of neoclassical cost function, both of which determine the sector's price level as a function of costs. Alternatively, there may be a freely varying sectoral output price that equilibrates supply and demand and also acts as a vehicle for forced saving at the macro level. In the latter case some component of cost (e.g., the markup rate) will have to be endogenous to allow the SAM column identity to remain in force when the model is perturbed.
3. To describe circular flow, the model also needs rules for mapping each sector's value flows into real demands for output from itself and other sectors. Examples include input-output coefficients, and equations saying how saving is determined from each income flow (generated from one or more sectors, e.g., total wages) and how each nonsaved portion is distributed across demand by sector. Such a sectoral spending breakdown will have to satisfy adding-up restrictions as discussed in the preceding section.
4. Finally, there has to be a rule for each sector's quantity adjustment, obeying the SAM row identity stating that its value of output must equal the sum of values of components of demand. For example, output may be determined by demand only. Alternatively, there may be a sector supply restriction, in which case some other quantity variable will have to adjust to bring sectoral demand into equality with supply. Price-sensitive compo-

nents of demand provide one such vehicle—as we will see shortly a combination of forced saving and consumer expenditure responses may be involved. If prices are determined from the supply side, trade flows or state purchases are obvious candidates to be adjusting variables in some sectors, as discussed in sections 1.5 and 1.6.

Besides these within-sector specifications, cross-sectoral relationships also apply. The most important stems from the savings leakages that are built into all applied CGE models. With saving present in the system, there must be a demand injection in at least one sector to permit a sensible solution to exist. The simplest Keynesian model (with P normalized to unity) illustrates. The four sets of rules just discussed take the following form:

1. Income equals the value of output, or $Y = PX$.
2. Prices are normalized, or $P = 1$.
3. Consumption is a fraction of real income, or $C = (1 - s)Y/P$.
4. Output adjusts to satisfy demand-supply balance, that is, $X = C + A$.

It is clear that a nontrivial solution cannot exist without a nonzero demand injection A . A variable like A may be a function of other variables in the system, for example, a contemporaneous investment accelerator $I = I(X)$ or an export demand $E = E(P)$, but it cannot be determined identically from income flows. This general argument carries over to other adjustment mechanisms and many sectors.

Besides demand injections one may also choose to impose cross-sectoral supply constraints. Obvious examples are supply functions for total labor or specific skill classes, or a binding trade balance. Each additional restriction will require a corresponding endogenous variable. We have seen in section 1.5 how a labor supply constraint can be met by forced saving or an endogenous nominal wage (in which case institutional factors in wage formation have to be omitted). Similarly section 1.6 shows how a binding trade constraint again requires forced saving, or perhaps an endogenous nominal exchange rate.

Any number of examples could be concocted to illustrate these general observations. To wind up this section, we content ourselves with just one. It underlies several models in this book as well as an influential investigation of the agricultural/nonagricultural terms of trade (deJanvry and Sadoulet 1987). The two-sector specification blends forced saving, Engel demand effects, and output adjustment in an illuminating fashion. The equation set is simplified from table 1.5 and appears in table 1.6.

We maintain the identification of sector 1 with "agriculture" and sector 2 with the rest of the economy, or "industry." Agricultural supply X_1 responds to the price P_1 in the first equation, presumably with a low elasticity. Equation (2) shows that P_2 is determined by a markup over labor costs. Equations (3) through (5) determine income flows, and (6) deducts each class's saving from income to determine total consumption demand D .

Although they have different saving rates, we save symbols and space by assuming that consumer behavior of the groups is the same (ignoring the aggregation error that results from not using separate demand functions for each). On this hypothesis, (7) and (8) describe consumer demand. The equations are simplified from the linear expenditure system described in section 1.11. They show first that adding-up is assured, that is, $P_1 C_1 + P_2 C_2 = D$. Second, it is easy to verify that the elasticity of C_1 with respect to D is $\alpha D / (\alpha D + BP_1)$ which is less than one for $\theta > 0$. Equations (7) and (8) provide an easy way to assure that Engel's law for consumption of agricultural commodities is satisfied.

Equation (9) is an excess demand function for agriculture. With the supply function (1) imposed, P_1 must vary to bring demand in line with supply. Demand components are exports E_1 and consumption C_1 . Since industry's price is determined by a markup from (2), its output X_2 adjusts to satisfy the excess demand equation (10). If industry is the only sector supplying capital goods, its demand is made up of investment by destination $I_1 + I_2$, plus consumption C_2 . The investment-saving balance (11) can be derived from (1) through (10). It is always a good check on the data and accounting consistency of an applied model, but is not an independent equation.

We have a simple fix-price/flex-price specification, combining price-sensitive supply and demand, forced saving, and output adjustment. To see how the three mechanisms mix, we can boil the model down to excess demand functions for agriculture,

$$\frac{\alpha \gamma_2 P_2 X_2}{P_1} - (1 - \alpha \gamma_1) X_1(P_1) + \theta + E_1 = 0,$$

and Industry,

$$I_1 + I_2 - [1 - (1 - \alpha) \gamma_2] X_2 + \frac{[(1 - \alpha) \gamma_1 X_1(P_1) - \theta] P_1}{P_2} = 0,$$

in which the γ_i stand for consumer spending from sectoral output flows $P_i X_i$, and are defined in table 1.6.

Table 1.6
Macro relationships for a simplified two-sector model

Structural relationships	
$X_1 = X_1(P_1)$	(1)
$P_2 = (1 - \tau_2)\omega_2 b_2$	(2)
$Y_a = P_1 X_1$	(3)
$Y_w = w_2 b_2 X_2$	(4)
$Y_r = \tau_2 w_2 b_2 X_2$	(5)
$D = (1 - s_a)Y_a + (1 - s_w)Y_w + (1 - s_r)Y_r$	(6)
$c_1 = [\alpha D + \theta P_1]/P_1$	(7)
$C_2 = [(1 - \alpha)D - \theta P_1]/P_2$	(8)
$E_1 + C_1 - X_1 = 0$	(9)
$(I_1 + I_2) + C_2 - X_2 = 0$	(10)
Investment-saving balance	
$P_1 E_1 + P_2(I_1 + I_2) - (s_a Y_a + s_w Y_w + s_r Y_r) = 0$	(11)
Definitions	
$\gamma_1 = 1 - s_a$	
$\gamma_2 = [(1 - s_w) + (1 - s_r)\tau]/(1 + \tau)$	

The second equation determines X_2 as a function of P_1 and other variables. Greater investment demand or an upward shift in the agricultural supply function makes X_2 rise. On the other hand, a higher P_1 will make X_2 decline if the supply response to a higher price is weak and θ is large. The latter condition is tantamount to a low income elasticity of demand for agriculture. The story is that a higher P_1 reduces real wage income, leading to forced saving. With income-inelastic sector 1 demand, “automatic lack-ing” largely cuts demand for industrial goods, and X_2 falls. Alternatively, X_2 could rise if agricultural supply is elastic, farmers account for a big proportion of demand, or Engel effects are weak.

The first equation is simpler. It simply shows that higher exports or sector output will raise demand for sector one products, causing the price P_1 to rise. The two equations together determine a macroeconomic equilibrium based upon the agricultural price and industrial output.

The obvious comparative static experiments are to raise agricultural exports (or assume an adverse agricultural supply shock) and investment demand. The former drives up P_1 . The subsequent effect on demand may make industrial output either rise or fall. Higher investment increases industrial output; the induced demand response causes agricultural price inflation. Outcomes of these combinations of forced saving, output

changes, and shifts in demand composition are important for policy formation in developing countries, as illustrated in several models in this book.

Finally, note that the fix-price/flex-price specification carries over to a number of other areas of interest, such as traded goods with prices fixed from abroad as opposed to *nontraded goods with market-clearing* prices. This so-called dependent economy model can be given a structuralist interpretation (Taylor 1983) and also underlies a number of more neoclassical CGE models of trade policy surveyed by deMelo (1988).

1.9 Financial Flows and Stocks

So far, the models (passing remarks about real balances included) have addressed only the part of the economy that produces goods and services. Most CGE work has been devoted to this IS side of the system, but computable models can deal with LM issues as well." A sample data base for real-financial models is set out in this section. The one following takes up modeling questions per se.

Table 1.7 is a SAM with a compressed but illustrative set of financial stocks and flows. It builds upon the matrixes already presented (especially in being set out in nominal terms) and extends them in three ways. First, the new SAM shows how saving by each class or institution feeds into changes in asset and liability positions in current prices. Second, financial flows are cumulated into stocks that fit into a portfolio balance model. Finally, the matrix helps one visualize channels for interaction between the real and financial sides of the economy.

The real side is described by rows 6 through 10 and columns 1 through 6. Value added is broken down into income flows to firms (V_f), households (V_h), and the government (V_g) in rows 7 through 9 of column 1. In row 10 foreigners receive income from imports of intermediates and capital goods. Row 6 shows components of demand. Columns 2 through 5 give uses of incomes, and column 6 shows national (row 6) and imported (row 10) provision of investment goods by origin. Government revenue in row 9 comes from value added (indirect taxes and public enterprise profits) and direct taxes on firms (T_f) and households (T_h).

The first big difference between table 1.7 and the SAM's presented heretofore is that savings flows from the different income categories are shown *in* separate rows—11 for firms, 12 *for* households, *and so on*. Across a row, saving goes into net accumulation of assets by the corresponding income class. The sign convention in these flows of funds is that

Table 1.7
Illustrative symbolic SAM combining the real and financial parts of the economy

	Production cost 1	Uses of incomes					Firm portfolio				Government portfolio				Bank portfolio				Totals 18
		Household		Government	Foreign	Capital formation	Physical capital	Bank borrowing	Foreign borrowing	Household portfolio deposits	Bank loans	Foreign loans	Foreign portfolio Reserves	Loans	Reserves	Deposits			
		Firm hold																	
		2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17		
initial portfolio																			
1 Capital							$-P_i \bar{K}$										$-P_i \bar{K}$		
2 Bank loan								$\bar{L}_f$		$\bar{L}_g$				$-\bar{L}$			0		
3 Foreign loan									$e\bar{L}_f^*$		$e\bar{L}_g^*$	$-e\bar{L}^*$					0		
4 Reserves													$e\bar{R}^*$		$-e\bar{R}_g^*$		0		
5 Deposits										$-\bar{H}_g$						$\bar{H}$	0		
Sales																			
6 Product	$Pq_i X$		PC	PG	PE	$P\beta_i I$											PX		
Incomes																			
7 Firm	V_f																		
8 Household	V_h																		
9 Government	V_g	T_f															Y_g		
10 Foreign	$eP_g^* a_g X$						$eP_f^* \beta_o I$										Y_{for}		
Flows of funds																			
11 Firm		S_f					$-P_i I$	ΔL_f	$e\Delta L_f^*$										
12 Household																			
13 Government										ΔL_g	$e\Delta L_g^*$								
14 Foreign													$-e\Delta L_f^* + e\Delta R^*$						
15 Bank							S_{for}								$-AL$	$-e\Delta R_g^*$	ΔH		

Table 1.7 (continued)

[illegible]

Structuralist CGE Models

increases in assets are negative and increases in liabilities positive. Thus row 11 can be stated as

$$S_f + \Delta L_f + e\Delta L_f^* - P_f I = 0.$$

Investment by firms ($P_f I$) is financed by their own saving (S_f), increased loans from the banking system (ΔL_f), and increased foreign currency loans converted to domestic price terms by the exchange rate ($e\Delta L_f^*$). The difference operator Δ refers to changes between the end of last period (e.g., a year) and the end of this one: $\Delta L_f = L_f(t) - L_f(t-1)$.

A simple, though not atypical, financial structure is assumed. In row 11 household saving goes entirely toward building up net assets (AH_f) with the banks. In practice, households in any given period run up both bank loans and balances, but as a shortcut here their net asset increase is called new "deposits."

Banks in turn engage in new lending (ΔL_f) and build up foreign reserves ($e\Delta R_f^*$) to offset their higher deposits (AH_f) in row 15. The banking system's net saving is set to zero (or accumulated in firms's accounts), so there is no saving entry toward the left end of its flow of funds row.

We have already seen in row 11 that firms take loans from both the domestic banking system and foreign sources. In row 13 the government does the same. Since the row sum must be zero, one of its entries has to be negative. Either the government is paying off loans or no great surprise—its current outlays (PG) exceed revenues (Y_g) in column 4 so that $S_g < 0$.

The final flow of funds is the balance of payments in row 14, stating that

$$e\Delta R^* = e\Delta L^* - S_{for}.$$

The net increase in reserves equals capital inflows less the current account deficit.

The next step is to add these flows to existing asset and liability stocks to get current positions. The time convention is that the asset and liability balances in rows 16 through 19 refer to the end of the current period and are evaluated in end-of-period prices where relevant. Stocks coming into the period are identified by overbars and appear in rows 1 through 5. They are also valued at end-of-period prices. Thus capital gains between periods have already been taken into account. Difficult valuation problems are swept under the rug by these conventions, but they substantially ease applied modeling. For a recital of the difficulties, see Foley (1975).

Reading down columns 7 through 17 shows how assets accumulate. For firms's capital stocks, for example, column 7 shows that

$$P_t K = P_t I + P_t \bar{K},$$

or end-of-period capital equals investment plus capital at the beginning of the period (for simplicity, depreciation is ignored). Using explicit time arguments, this equation can be restated as

$$P_t(t)K(t) = P_t(t)I(t) + P_t(t-1)K(t-1) + [P_t(t) - P_t(t-1)]K(t-1),$$

where $K(t)$ and $P_t(t)$ are the capital stock and its asset price at the end of the period, $I(t)$ is investment during the period, and $K(t-1)$ and $P_t(t-1)$ are the stock and asset price at the end of last period or the beginning of this one. The capital gain term $[P_t(t) - P_t(t-1)]K(t-1)$ accrues over the current period.

The capital stock is carried down to row 20 ("totals") since it is difficult to model the market for old buildings and machines. Otherwise, end-of-period stocks appear in the asset balance rows 16 through 19. In row 16 outstanding bank loans (L) go to firms (L_f) and the government (L_g). Foreign loans (eL^*) are taken by the same two borrowers in row 17. By the signs in table 1.7, both these balances can be interpreted as excess loan supply equations. Similarly, the supply of foreign reserves (eR^*) is held by banks (eR_b^*) in row 18. Bank deposits (H) are equal to the amount supplied by households (H_h) in the last asset account.

These ending balances are stated for beginning-of-period stocks in rows 2 through 5 (with signs reversed). It is illuminating to sum the flows of funds rows 11 through 15, and then insert the initial asset balances. The outcomes are the *equalities*

$$\begin{aligned} P_t I - (S_f + S_h + S_g + S_{\text{tot}}) &= 0 \\ &= (L_f + L_g - L) + e(L_f^* + L_g^* - L^*) \\ &\quad + (H - H_h) + e(R^* - R_b^*). \end{aligned}$$

In the first line we already know that investment will equal saving from the real side of the model. The second line states that macro balance implies that the sum of excess asset demands must be zero. In equilibrium therefore only three out of four (or $N-1$ out of N) of the asset balances represent independent restrictions on the system. How they interact with the real side of the model is the topic we take up next.

1.10 Real-Financial Models

A description of all the models that can be built around the accounts in table 1.7 is out of the question—that is what textbooks in macroeconomics

are about. Rather, we will take up formulations appropriate to the developing country context, starting with complete systems and then going on to models tailored to fit available financial data (chiefly of the monetary sort).

Table 1.7 is best read top to bottom, or north to south. We start the discussion from a consistent real side solution (with only one sector, for simplicity) and the saving flows it generates toward the west of the matrix. Available savings put restrictions on the changes in assets and liabilities that appear in rows 11 through 15. These changes together with initial stocks in the northeast feed into the closing asset balances in the southeast. The first task is to describe how asset markets clear. The effects of financial market outcomes on price and quantity adjustment is the next topic, reading the matrix from southeast back to west and northwest.

A rather Keynesian story can be told along the lines suggested by Tobin (1982). Suppose that output is the variable that adjusts on the real side, with either markup or neoclassical pricing. At a price-quantity equilibrium, there will be a government saving flow (possibly negative) S_g . If the government is subject to an external credit limit, it will be able to find “new money” in a fixed amount ΔL_g^* from abroad. Its new domestic borrowing AL_g , follows from row 13 of table 1.7, and its total loan obligations L_g and L_g^* from columns 11 and 12.

Next, in row 12 household saving supply S_h generates an increment in desired deposits ΔH , and thereby total deposits H_h from column 10. These are accepted by banks in row 19, determining the money supply H . We suppose that banks choose to lend out some fraction $\lambda(i, e)$ of their deposit base, where i is a market-clearing loan rate of interest and e is included to reflect the attractiveness of holding reserves. Loan supply to firms comes from row 16,

$$L_f^s = \lambda(i, e)H - L_g.$$

Firms's loan requirements depend on their initial stocks together with their flows of funds in row 11,

$$L_f + eL_f^* = \bar{L}_f + e\bar{L}_f^* + P_i I \quad S_f = Q_f.$$

It is reasonable to postulate that firms borrow more domestically when the interest rate is low or the exchange rate high:

$$L_f^d = \delta(i, e)Q_f$$

The excess demand equation, $L_f^d - L_f^s = 0$, solves for the interest rate i and level of bank loans L_f . Foreign loans L_f^* are determined by the firms's

initial asset position and their flows of funds. The asset balances in rows 16 and 17 sum outstanding loan stocks to get the bank and foreign totals L and L^* .

Now we are virtually done. Initial assets and the foreign flow of funds in row 14 determine reserves held abroad R^* . The quantity R_b^* that banks wish to hold comes from initial positions, the household and banking system flows of funds in rows 12 and 15 and the deposit balance in row 19. These two variables will take the same value, in accord with our observation last section that there is no need to impose the reserve balance in row 18 as an independent restriction on the system.

These manipulations basically produce portfolio allocations and a rate of interest i , which can feed back to the real side. The interest rate will be affected by both incoming asset and liability stocks and current financial flows, in line with the market's conventional wisdom. The influence of commonly used instruments of monetary policy—rediscount, reserve requirements, credit controls—could be modeled by separating the accounts of the central and commercial banks which are consolidated in table 1.7. If there were a market in government securities used as an arena for open market operations, it would be easy to include. Finally, the exchange rate will affect loan market equilibrium via both demand and supply.

In the model just sketched, the interest rate is the main variable mediating effects of finance on output and prices. Three channels are often specified. First, net interest payments should be included in income statements. Such flows are often ignored as being small, but this is not always the case.

Second, investment demand is presumably a negative function of the nominal (i) or real ($i - \dot{P}$) interest rate—this is the traditional LM channel. **It may be supplemented by positive effects of the interest rate on savings** rates (or on the flow incomes of high-saving households), if there is reason to believe that strong responses exist.

Third, there is evidence in some economies that interest rates affect prices directly by raising the cost of financing working capital. If a fraction ξ of variable costs BX must be financed by bank loans, then the markup equation may take the form

$$P = (1 + \tau)(1 + i\xi)B,$$

so that interest rate increases drive up the price level. Loans demanded by firms may also be tied directly to their working capital requirements as opposed to their overall financial deficit. Both specifications have been used in applied models.

Comparative statics in the Tobin-style model go along conventional lines. Expansionary fiscal policy is associated with greater government credit needs and commercial bank borrowing. Deposits also rise, increasing loan supply. Disequilibrium in the loan market is eliminated by interest rate adjustments (presumably increases) that then influence investment demand and the price level. The less flexible is the interest rate, the greater is the **part of the expansionary impetus that will spill abroad in the form of reduced bank reserves and higher foreign borrowing by firms.**¹²

More monetarist-sounding outcomes could also be arranged. With labor **supply predetermined**, the real side would have to be closed by forced saving and/or real balance effects as discussed in sections 1.4 and 1.5. Fiscal expansion would cause inflation as well as increased borrowing from the **banks. Nominal savings flows from rentier incomes would also rise**, increasing the end-of-period deposit base. With appropriate parameters, rough proportionality between the money supply and price increases would be observed, “ratifying” the equation of exchange. Results would go in the same direction even with variable employment, via a neoclassical cost function or an assumed positive elasticity of the markup rate with respect to the level of capacity use.

Several other effects can be built into applied models, and bear mention in passing. First, fixing the sum of changes in reserves and total foreign borrowing would amount to a foreign exchange constraint. Its effects could be modeled by endogenizing the exchange rate or assuming forced saving adjustment on the real side, as discussed in section 1.6. Expansionary policy would lead to price and interest rate increases.

Second, the interest rate might be controlled. There would have to be some sort of quantity adjustment in financial markets. Government foreign borrowing and/or the banking system’s foreign exchange reserves could be made endogenous.

Finally, asset valuation issues have to be considered. Devaluation would increase the borrowing requirement of firms Q_f (defined above) by appreciating initial foreign liabilities $e\bar{L}_f^*$ in local currency terms. The outcomes could involve drops in investment for reasons of reduced cash flow, interest rate increases, and stagflation. Contrariwise, foreign asset positions would be enhanced by devaluation. If rentier or firm-owning households have substantial holdings abroad (the fruits, perhaps, of previous capital flight), devaluation would make them wealthier in local currency terms. Their saving rates might decline, with expansionary consequences.

In closing, a word should be added on incomplete asset market specifications. One important version uses only the flow of funds 13 through 15 in

table 1.7—private sector net savings surpluses are left out. Such a model hinges around determination of the money supply by government borrowing and the balance of payments, and its effects on prices and quantities. Depending on the assumptions one makes, either monetarist or structuralist outcome can appear.

The implication is that even when full information on financial stocks and flows is not available, it is still feasible to incorporate monetary considerations into CGE models. Whether they are of sufficient importance to justify the labor involved is of course a matter for the model builder to judge. The fact that the International Monetary Fund often bases its stabilization programs on a rudimentary version of table 1.7 closed with the equation of exchange and forced saving or Meade/Johansen rules (Taylor 1988a) suggests that alternative specifications might usefully be thrown into the policy debate.

1.11 Nonproportionalities and Substitution

In this section functional specifications for applied models are quickly reviewed—the use of intercept terms, constant elasticity functions, input substitution as captured by CES functions, and consumer demand behavior in the LES specification. All these tricks are used extensively in the models of this book. The section closes with observations about the use of time series information to specify parameters.

Intercept Terms

All the ratio estimates of parameters discussed in section 1.3 can easily be extended to include intercepts. For example, suppose that wage income and saving from it take respective values of Y_w and S_w in the base year SAM. One can easily estimate an average (= marginal) saving rate s_w as the ratio $s_w = S_w/Y_w$. Alternatively, one might believe that the marginal saving rate differs from the average so that the saving function is $S_w = s_w Y_w + \bar{s}_w$. The intercept term $\bar{s}_w$ can be computed from the SAM data base if one assumes a value for the marginal rate s_w (e.g., obtained from consumer budget surveys or a time series regression). $\bar{s}_w$ will be an elastic (inelastic) function of Y_w when $\bar{s}_w$ is negative (positive). This intercept trick can be extended to more general systems in which one wants to avoid purely proportional (or unit elastic) relationships. It is the secret of the linear expenditure system of consumer demand equations discussed below, for example.

Intercepts often prove useful in saving, investment, and export functions. They have the advantage that the intercept terms can be manipulated to take into account intuitive or exogenous information (supply shocks, likely changes in behavior, political events) about how the economy is behaving. **If I take the intercepts and you take the slopes, I'll be better able to tune a model solution to fit my views about the current economic situation than you.**

Constant Elasticities

This same observation applies to constant elasticity functions, for example, $E = \bar{E}[eP^*/(1 - z_e)P]^\eta$ for exports in the notation of table 1.2. The constant term $\bar{E}$ can be calculated from the base-year SAM once one knows the export level E , subsidy rate z_e , the price terms, and postulates the elasticity η . Like the linear functions just discussed, constant elasticity formulations are *convenient to use* when an isolated quantity variable responds to a price and vice versa. Export functions, markup rates (which may depend on the level of activity or monetary variables), money demand functions, import *coefficients*, and *sectoral* supply functions making output a function of the price-cost ratio (or $1 + \tau$) are all examples. Economists often have stylized elasticities for such relationships in *their* heads, and they are easy to plug into a model.

CES Functions

One often wants to specify substitution possibilities between production inputs—capital and labor, imported and domestically produced intermediate inputs, oil and gas as energy sources. A load of distributional baggage is carried along with any such neoclassical specification, as we have seen in section 1.4. Nonetheless, where substitution is deemed to be potentially important, it should be included in a CGE model. Constant elasticity of substitution (or CES) functions are often convenient, for reasons to be indicated shortly.

A CES cost function (e.g., for labor L and capital K) takes the form

$$P = [\alpha_L w^{1-\sigma} + \alpha_K (rP)^{1-\sigma}]^{1/(1-\sigma)},$$

in a one-sector model in which w is the wage rate, P the output price level, and r the rate of profit. The constant elasticity of substitution is σ .

Applying Shepard's lemma (see section 1.4) shows that input demand functions take the form

$$\frac{L}{X} = \alpha_L \left(\frac{w}{P} \right)^{-\sigma},$$

and

$$\frac{K}{X} = \alpha_K (r)^{-\sigma},$$

where X is the level of output. These equations have the advantage of making input-output ratios depend *only* on the cost variable relevant to the input at hand—the real wage for labor and the profit rate for capital. Cross-cost terms do not appear, as they do in the general neoclassical input demand functions in table 1.3.

This feature of CES functions makes them easy to interpret and manipulate. “Distribution” or scale parameters like α_L can be calculated readily from a SAM once one specifies the elasticity of substitution σ , since the labor share of output is equal to $\alpha_L (w/P)^{1-\sigma}$. The functions also “nest” nicely. For example, aggregate labor cost w could be constructed as a CES index of wages w_i of different skill classes L_i . Employment of each skill group (L_i/X) would depend only on the cost ratios w_i/w and w/P .

Besides input substitution harmonic means of the CES type can also be used to indicate trade-offs such as the one between domestic and imported goods in satisfying intermediate requirements (see section 1.6), or between domestic and export destinations of the same sector’s products in a constant elasticity of transformation (CET) setup. These formulations are often adopted in World Bank sponsored CGE models (Robinson 1989) and add a degree of price responsiveness to the competitive versus noncompetitive trade categories used in most **models** in this book.

LES Demand Specifications

It would seem natural to carry the CES framework over to price substitution in consumer demand. However, a glance at the labor and capital input demand functions shown above reveals a major drawback—the functions are **“homothetic,” displaying unitary elasticities of inputs such as L with respect to output X .** Since income elasticities of demand for most goods differ from unity (recall Engel’s law), one needs a more general formulation. Applied CGE models often incorporate the linear expenditure system or LES.

The LES is derived in the usual way by maximization of a specific utility function,

$$U = \sum_{i=0}^N \mu_i \log(C_i - \theta_i),$$

subject to the budget constraint,

$$\sum_{i=0}^N P_i C_i = D.$$

A bit of manipulation shows that the demand functions coming from this exercise take the convenient linear form

$$C_i = \theta_i + \frac{\mu_i}{D - F} (D - F),$$

where

$$F = \sum_{i=0}^N P_i \theta_i.$$

The interpretation of these equations goes as follows: The μ_i parameters are “marginal budget shares,” which can be estimated as

$$\mu_i = \eta_i \left(\frac{P_i C_i}{D} \right) - \eta_i \psi_i,$$

where η_i stands for the income elasticity of demand for commodity i and ψ_i is its budget share in the spending of the relevant consumer class from the base year SAM.

The θ_i parameters are fixed levels of consumption of each commodity, independent of income or price. The quantity F is the value of a “floor” consumption basket defined by the θ_i and the current set of prices P_i . Any excess of total spending D over the floor is directed to commodities in line with their marginal budget shares μ_i . If all the θ_i were zero, then all commodities would have constant budget shares, implying unitary income elasticities of demand. These intercept terms permit the LES to produce income elasticities differing from one.

Further algebra shows that the θ_i must satisfy the condition

$$\theta_i = \left(\frac{D}{P_i} \right) (\psi_i - \xi \mu_i),$$

where

$$\xi = \frac{D - F}{D}$$

The new parameter ξ is usually called the “supernumerary income ratio” because it measures the amount of available spending power that consumers have above the floor level F . A larger value for ξ indicates that consumer behavior will show more sensitivity to changes in income and prices.

The foregoing discussion shows that the equations of an LES for $N + 1$ commodities (domestic goods plus imports) can be fully parameterized from $N + 1$ budget shares ψ_i (which come from the base-year SAM), $N + 1$ income elasticities η_i , and the parameter ξ . The income elasticities and ξ can often be inferred from household budget data and/or time series information. Absent any other knowledge, the stylized fact is that ξ often takes a value near 0.5 (Lluch, Powell, and Williams 1977).

These same data also serve to parameterize any other consumer demand system derived from an additive utility function of the form

$$U = \sum_{i=0}^N f_i(C_i).$$

If the f_i are exponential functions (same exponent but different constant terms for each good), one gets a “direct addilog” set of demand equations that come close to having constant income and own-price elasticities. For the details, see Taylor (1979, app. E).

Using Time Series Information

All the manipulations we have discussed boil down to making functions with specified slope or curvature parameters (elasticities, marginal rates or shares, etc.) consistent with the base-year SAM by adjusting their scaling parameters or constant terms.

The reason one wants to do this sort of “calibration” is to ensure that solutions for a CGE model take off from an initial position that incorporates all the consistency conditions that its accounting and behavioral equations contain. Consistency and clear causal links are the main strengths of the CGE approach.

Time series information can be used to specify slope and curvature parameters for CGEs—all one has to do is rescale the regression intercept terms so that the regression line passes through the SAM data points instead of the statistical sample’s mean. Whether this is advisable in whole-sale fashion depends on three considerations.

First, data over time are usually not available for the interindustry flows, origin-to-destination investment flows, consumer budget breakdowns, and

so on, that figure in applied CGEs. Model builders are necessarily driven toward more ad hoc calibration procedures based on partial information.

Second, economies (especially developing economies) change. It is never clear what numbers from even ten years ago have to do with the current economic situation, especially when model closures or directions of macroeconomic causality may be very different. Relying on regression equations which amount to averages of past years's data with equal weights may not be an illuminating procedure. However, for key relationships such as investment demand functions and wage indexation rules, incorporating time series information often makes sense. Fully wedded CGE and econometric models are probably impossible, but both parties may be happier if they flirt a little bit.

Third, one may try to use theory to express SAM calibrations in terms of "deeper" parameters to be estimated cross-sectionally or over time. Neoclassical economists are fond of setting up optimization models to this end. When cleverly done, such formulations have intuitive content, as the CES and LES equations illustrate. They also have their drawbacks. There is no clear evidence that "agents" such as households, firms, or the state truly optimize. Memory is a good test, and anyone who has ever balanced a home's or enterprise's books or participated in IMF negotiations in full tilt has vivid recollection of groping, satisficing, irrationality in the common usage of the word, and worse. Moreover, although setting up and solving simple optimizations is an interesting game, it can easily divert attention from fundamental questions of political economy, macroeconomic causality and class. Making sensible judgments about slopes, intercepts, and closure is far more crucial than incorporating "deep" (and ultimately unobservable) parameters to an applied model's success.

1.12 Dynamic Relationships

It is straightforward to simulate CGE models over a number of periods, to get projections of possible future behavior of the economy. These solutions are conditioned by the assumptions one makes about comparative static **properties of the model, as well as exogenous time trends. If these dynamic** exercises are carefully thought through and executed, they can provide useful information.

Obviously, any parameter in a model can be given a time index and a trend. In addition many variables have to be updated from period to period in simple ways. The main examples of both types go as follows:

1. Asset and liability stocks should be cumulated from within-period flows along the lines indicated in table 1.7. If δ_i is the physical depreciation rate for the capital stock K_i in sector i , the accumulation rule will be

$$K_i = I_i + (1 - \delta_i)K_i,$$

where I_i is the sector's gross capital formation by destination. Parameters like δ_i can be estimated as ratios from depreciation flows in a SAM and initial capital stock estimates (or addivinations).

2. Productivity increases are typically observed in time series output data. They are usually not well understood in terms of explanatory factors, but at least can be modeled by attaching trends to the relevant production input parameters. Growing labor productivity, for example, would be reflected by a steady geometric decrease in the labor input coefficient b in a markup formulation or the parameter α_L in the CES labor demand function discussed last section. Time trends in consumption can be captured by attaching growth rates to the floor parameters θ_i in an LES specification, subject to the adding-up requirement that

$$\sum_{i=0}^N [\theta_i(t) - \theta_i(t-1)] = 0.$$

3. Predetermined nominal prices may have to be updated from one period to the next in line with inflation and other economic variables, for example, by a rule such as

$$w(t+1) = w(t) \left\{ 1 + \varepsilon \hat{P}(t) + \alpha \left[\frac{L(t) - \bar{L}(t)}{\bar{L}(t)} \right] + \beta \left[\frac{\bar{\omega}(t) - \omega(t)}{\bar{\omega}(t)} \right] \right\}.$$

Here, wages next period are indexed to the current periods price inflation (with a “pass-through coefficient” ε which usually takes a value between zero and one). The between-period wage increase also depends on employment L relative to a “full” level $\bar{L}$ in a Phillips curve specification as well as distributional tension as measured by the shortfall of the real wage ω ($= w/P$) from a target level $\bar{\omega}$. Such relationships underlie cost-based structural inflation and are used in several models in this book.

4. Similar considerations apply to other price variables such as the exchange rate e . It may be indexed to ongoing inflation in a “crawling peg” and also adjusted between periods in light of the balance of payments results from the model.

5. These trending price variables can also have feedbacks to the rest of the system. The inflation tax and other effects of price inflation $\hat{P}$ as discussed

in section 1.5 are prominent examples. Also unless savings rates and stocks of assets held by different classes are broadly consistent, dynamic simulations can give unstable wealth distributions unless there are improbably large relative shifts in asset prices.

6. Various exogenous items have to be specified—government spending (in real or nominal terms), external and internal credit limits, investment levels by destination and export and import flows (if not determined by closure or behavioral relationships), and inventory changes are the obvious examples. It is often convenient to set up their changes over time in growth rate form.

1.13 Solution Techniques

CGE models aren't terribly useful unless they can be solved numerically in a reasonable amount of time. Computer speeds go up by about a factor of ten every decade, so that by the late 1980s solving a model with up to a few dozen sectors is no major task (though the same cannot be said about putting its data together). Despite computer speed one does have to think a bit about setting up solution algorithms. Basically, four approaches appear in the literature (Robinson 1989).

First, in each period the model boils down to a set of nonlinear equations with a certain structure, around which solution techniques can be devised. Details for models like the ones in this book are set out below. Such model-specific procedures are often convenient (and help one think about how the model works) but they are not essential. Nonlinear algorithms on mainframes (or even spreadsheet packages on microcomputers) now deal easily with more than 100 equations. That is enough for a model with quite a few sectors

Second, models in the Arrow-Debreu tradition without forced saving and other nonhomogeneous consequences of nominal flows can be solved by finding fixed points of their excess demand functions in nonnegative prices. The Scarf fixed-point algorithm provided the original impetus for neoclassical models for industrialized countries (see, e.g., Scarf and Shoven 1983; Shoven and Whalley 1984). This approach is not of much use for structuralist models. In practice, even neoclassical modelers have given up fixed-point algorithms of late in favor of standard nonlinear equation-solving computer routines.

Third, more or less neoclassical models can be set up as nonlinear programs instead of fixed-point problems. Ginsburgh and Waelbroeck (1981) provide examples and a survey of techniques.

Finally, all the equations can be linearized around the base-year SAM so that an approximate solution can be found by matrix inversion. Johansen (1960) adopted this technique for his pioneering CGE model, and it has been used widely since, such as in the large models for Australia designed by Dixon, et al (1982).

Models of the type in this book can usually be solved by simple iterative procedures based on the circular flow of incomes discussed in section 1.8. Consider a model in which output or some other quantity variable (a trade flow or real government spending) adjusts to ensure demand-supply balance by sector. Each sector's price can be formed by a markup rule, neoclassical cost function, or the inverse of a supply function depending on the price-cost ratio.

One starts with initial (or "old") guesses at sector quantity variables. Through the equations **corresponding to columns of the SAM, they permit** computation of sectoral prices and income flows. These in turn feed through saving functions and complete demand systems to give real consumption levels. Together with sectoral demand injections (exports, investment, and/or government spending) the consumption flows provide "new" estimates of quantities through the input-output balance equations. One proceeds in this fashion comparing new with old quantity levels until they agree within a pre-set tolerance level. This procedure usually converges after a couple of dozen iterations with the sorts of savings leakages **that** appear in applied models.

One or more sectors may be characterized by flexible prices, for example, the overall price level when there is macroeconomic forced saving, the agricultural price in the model of section 1.8, or the interest rate for the loan market in section 1.10. Each such price is determined implicitly at equilibrium by the condition that its corresponding excess demand should equal zero. The question is how to fit price adjustment into an iterative procedure like the **one** just discussed.

Depending on the model specification, one may be able to get an explicit expression for a flex-price by inverting its excess demand function—this could be used in conjunction with output levels computed as described above. Alternatively, **prices** could be iteratively modified parallel to outputs in proportion to excess demands-experimentation with factors of proportionality usually gives convergent solutions. Finally, outputs could be computed as above for given prices, while in an outer iteration loop the prices are adjusted by some nonlinear equation algorithm to drive the corresponding excess demands to zero. All these procedures have been used successfully in practice.

The last complication is that the model may be subject to cross-sectoral restrictions such as labor supply functions or a balance of payments constraint. It can always be solved as above for given values of the variables that are supposed to adjust to meet the constraints. In an outer solution loop **these variables can then be changed to drive their corresponding** constraints toward being satisfied. For a reasonable number of restrictions (e.g., up to ten), this procedure is perfectly feasible with standard nonlinear equation packages on mainframes. If one is willing to wait a bit for solutions, the same technique can be applied on microcomputers.

In closing, it is worth noting that solution procedures for applied models always seem to give unique results, perhaps because most model variants that one considers are perturbations around the initial solution represented by the base-year SAM. One uses the model first to replicate the SAM and then do comparative statics or conditional projections. Fortunately, neither exercise is plagued in practice by different solutions for the same set of exogenous variables and parameters.

1.14 Preview of the Models

The chapters that follow describe a dozen applied structuralist CGE models. They begin with three fix-price/flex-price models of the type discussed here in section 1.8, designed to deal with food subsidy, fiscal, and macro-economic stabilization issues in Mexico, Colombia, and Nicaragua. Chapters 5 through 7 extend this general line of analysis to effects of public enterprise pricing policy in India, energy pricing in Egypt, and subsidies and general macro policy in Sri Lanka.

Chapter 8 applies **several** "partial" CGE's to India, to ask how its growth prospects might be limited by constraints of several kinds; chapter 9 looks more in detail at different sorts of inflationary processes in that country. The next two chapters set out models with full financial sectors (monetarist linkages are already considered in other models) to explore impacts of wealth redistribution in Mexico and monetary policy in Thailand. Chapters 12 and 13, respectively, take up medium term dynamics **of productivity growth** and lagging oil revenue per capita in Kuwait, and inflation stabilization in Argentina.

In more detail, the models deal with the following issues:

The chapter 2 model by **Nora** Lustig and Lance Taylor was set up to address problems of food subsidy design in Mexico in the late 1970s. It illustrates the general nature of fix/flex models by comparing the effects on income distribution and the fiscal and external deficits of three types of

policies aimed at eliminating the food intake shortfall below a norm (as a proxy for a nutritional deficit) of the low-income population. The policies are a targeted food price subsidy, an income transfer, and a general price subsidy. The latter two are unacceptably costly in fiscal and foreign exchange terms, suggesting that effective targeting of subsidies would have to be a priority in any serious program aimed at alleviating malnutrition in Mexico.

Like the ones that follow, the Mexican model is multisectoral, with some sectors reaching market balance via price changes and others via output adjustment. Several income classes are considered, with rules specified for income generation and saving and consumption behavior of each class. Nominal wages and the exchange rate are fixed and macro causality runs from investment to saving, with both output shifts and forced saving processes contributing to the attainment of macro balance.

The model for Colombia in chapter 3 was put together by Juan Luis Londoño to deal with issues of fiscal and macro stabilization policy. Because they have differing spillover effects across sectors, the model distinguishes “general” and “specific” expenditure policies and distributional shifts. An example of a demand injection specific to a sector would be its exports, while spending from increased coffee growers’s income due to an export price bonanza would be a general distributional shift.

Londoño demonstrates the importance of terms-of-trade changes in macro adjustment and then goes on to consider the role of stabilization policy during three different episodes in recent Colombian history. The simulations provide useful insights, for example, how contractionary effects of devaluation in 1984–85 may have been partially offset by nonprice rationing of imports and a partial pass-through of the local currency price increase of coffee to the growers. The simulations also show that fiscal contraction is not a very effective means for reducing the external deficit in Colombia, while direct taxes (a “general” leakage) may be the best tool for cutting fiscal deficits.

Chapter 4, by Bill Gibson, takes up recent policy questions in Nicaragua. The model is foreign exchange constrained, with both internal output changes and forced saving serving as macro adjustment mechanisms. Gibson argues that recent macro policy in Nicaragua aimed at reallocating labor toward productive activities by reducing the forced saving which shifts the income distribution toward the informal sector which controls scarce imports makes sense. Three more controversial conclusions that follow from the logic of the model are (1) price controls may not be distributionally progressive, depending on their general equilibrium im-

pacts; (2) devaluation may prove progressive and expansionary, given the binding forex constraint; (3) a smaller trade deficit is **contractionary** and regressive, but attempts to shield the popular classes from these effects through nominal wage increases and higher government spending can **make the situation worse**.

Chapter 5 describes one of several CGE models for India that have been put together over the past decade by Hiren Sarkar and Manoj Panda (with inputs by **Jørn Rattso** and Lance Taylor). In the present paper Panda and Sarkar take up effects of price increases by public sector enterprises, which account for about a quarter of the nation's GDP. The overall price structure in the model is made sensitive to both cost pressure (e.g., for intermediates produced by public sector firms) and the level of aggregate demand. A detailed government accounting block of equations captures the effects of price changes on government revenue and spending, **which activate a chain** of macro effects.

The administered price sectors considered are petroleum, coal, electricity, and the railroads. The results first show that raising administered prices is stagflationary. However, when the resulting revenues are used to finance public investment, the outcomes are faster growth and inflation. On both counts, petroleum and coal price increases are more effective instruments for financing investment than are higher electricity or railroad rates. Finally, in a policy-relevant result, existing indirect taxes are shown to be better fund-raisers than administered prices.

The related issue of energy pricing by state-run oil refining and natural gas producing companies in Egypt is taken up by Nazli Choucri and Supriya Lahiri in chapter 6. They show, first, that if there is no substitution between oil and gas in the industrial sector and a completely elastic supply of gas, then a 10 percent oil product price increase will drive up various price indexes between **5** and **10** percent and lead to economic contraction unless state spending is raised by **8** percent.

The authors then extend the model in the neoclassical direction by permitting substitution between refined oil products and gas as intermediate industrial inputs (with an elasticity of substitution of **1.5**) and considering various supply elasticities for natural gas (infinity, two, and zero). In all of these cases when GDP is held constant by fiscal intervention, the elasticities of refinery output with respect to the oil product price range between about **-0.03** and -0.13 while elasticities of gas output range between zero and 0.27. The implication is that even when price-elastic substitution responses are built into the CGE model, its solutions suggest

that the practical possibilities for shifting between energy sources by altering their relative prices in the short run are relatively weak.

In chapter 7 Anne Maasland sets up a model to simulate the effects of an economic liberalization strategy pursued in Sri Lanka beginning in 1977. In 1980–81 the economy experienced rapid inflation due to cost-push from import price and wage increases, removal of price subsidies, and domestic money creation to finance ambitious agricultural investment projects. To capture these inflationary pressures, the model incorporates cost-push via markup pricing equations and also allows the markup rate to rise in response to increases in the money supply and capacity utilization, in keeping with the way the Sri Lankans view the inflationary process in their own economy.

The key policy issues center around income distribution, in a country where saving and consumption patterns differ between the poor and rich. An illustrative simulation maintains a food price subsidy to the poor and lowers public investment—it is counterfactual in both regards. The outcomes include lower prices, higher exports, and increased national consumption (especially of food by the poor). Increased tariffs on nonfood consumer goods imports can be used to pay for the subsidy and improve the current account which otherwise is adversely affected by consumer imports.

Chapter 8, by Lance Taylor, uses simple macro models that could be extended into full-fledged CGEs of the type presented in this book to try to isolate the binding constraints on the rate of growth of capacity (or potential output) in India. In a medium-term context the limits to growth from saving, productivity increases, foreign exchange, food supply, inflation from both monetarist and structuralist causes, and potential internal and external debt traps are considered.

Since they are parameterized over the same period in one economy, all the potential constraints are located near current Indian levels of the fiscal and trade deficits, and inflation and growth rates. In comparative dynamic analysis, they show that 6 instead of 5 percent capacity growth would involve widening the current account deficit or increasing the export share by 0.5 percent of the value of gross output (GDP plus intermediate imports), a 1 percent annual trend in the terms of trade toward agriculture, 2 or 3 percent faster inflation, and additional buildups of both external and internal debt.

These costs seem mild, suggesting that Indian policy has been conservative in the sense of keeping the economy demand constrained. The paper

concludes by discussing the political economy of various means to change this situation.

Chapter 9, by Jørn Rattso, goes into more detail in Indian inflation. Recent experience suggests that agricultural droughts and external terms of trade shocks are important sources of **increasing** prices. Conflicting claims among income groups act as propagation mechanisms to sustain inflations based upon these and other disturbances. The distribution of income among agriculturalists, wage earners, profit recipients, foreigners, and the state reflects the social structure. Five inflationary processes involving these groups-based on food prices, money wages, markups, import prices, **and** demand pressure from higher public investment-are investigated in a dynamic CGE.

The agricultural market emerges as the main ground on which macro-economic disturbances play. Government regulation of agricultural prices becomes a key anti-inflationary policy in the short run, while investment in the sector will be essential for price stability over time. Nominal wage **growth** and manufacturing profitability both turn out to be sensitive to food price inflation.

Chapter 10, by William Easterly, introduces a CGE model for **Mexico** which includes portfolio balances. These permit the analysis of important financial channels usually ignored in applied CGE work. Topics treated include (1) capital flight, which has been a major contributor to the Latin American debt crisis; (2) "dollarization" (or denomination of loans and deposits in dollars) of the local banking system; and (3) capital gains and losses on existing stocks of assets and liabilities due to price movements. With regard to the last topic, **the** inclusion of dollarized assets and liabilities broadens the usual analysis of real balances (see section 1.5) to cover **revaluations** induced by exchange rate changes. **For example, with** a local devaluation, the capital loss on dollar debt of firms has a negative effect on corporate investment in **the** model, while the capital gains on dollar assets of rentiers encourage their consumption. Redistributive effects of devaluation are complex, depending on the relative shares of different classes in dollar-denominated assets and liabilities, dollar expenditures, and dollar **incomes**.

Another novel feature of the model is incorporation of indexation of state-controlled prices and wages to endogenous price movements, introducing trade-offs for the government between its fiscal performance, the level of output, and distributional equity. Finally, the explicit inclusion of the banking system and the public sector accounts in the model permits assessment of the relationships between financial fragility and fiscal **perfor-**

mance. For example, the government could block a financial crisis which could be caused by revaluation of dollar-denominated domestic debts by specifying a lower exchange rate for such liabilities. But this would require heavy fiscal outlays to cover the losses suffered thereby by the banks. **The model** helps quantify, although not ameliorate, such serious policy problems.

Jeffrey Rosensweig and Lance Taylor take up other financial issues in a model for Thailand in chapter 11. Put together in 1984, it is one of the first CGEs to incorporate financial variables, and was in part designed to illustrate the sorts of policy analysis such formulations permit. The authors concentrate on short-run portfolio balances for households, firms, the government, and the commercial and central banks, asking how monetary policy affects macroeconomic equilibrium in a relatively open economy.

One policy query that arises is the extent to which different sorts of interventions are offset by reserve changes. If the government borrows 100 billion baht from the Bank of Thailand to increase public consumption, for example, that institution's reserve losses in the new macro equilibrium amount to 93—additional state spending is essentially financed from abroad. On the real side, the counterpart of the reserve loss is a large increase in imports since the Thai economy's marginal propensity to import (with respect to real GDP) is a relatively high 0.375. Central bank policy moves—changing the volume of rediscount or reserve requirements—are also at least partly offset by foreign reserve changes, reducing their impact on the interest rate and investment demand. Finally, devaluation has expansionary effects on the real side of the economy by stimulating exports. But when capital losses on external liabilities are taken into account, the expansion may well be canceled by increased interest rates and reduced investment demand.

Chapter 12 shifts focus to long-term development issues in Kuwait, a small and prosperous country which nonetheless faces vexing policy problems in the medium run. Motaz Khorshid develops a dynamic CGE model to quantify (1) the fiscal constraint that government is likely to face as its oil revenues per Kuwaiti (with 3.5 percent population growth) inevitably decline, (2) the social problem of trying to reduce the ratio of resident non-Kuwaitis to citizens from about 1.4 down to unity over a period of 10 to 15 years, and (3) diversifying the economy away from its extreme dependence on oil.

Two main strategies are considered. The first is reduction of aggregate demand to slow output growth and thereby employment of non-Kuwaitis, and the second is directed public investment to enhance productivity of

employed Kuwaitis. Productivity growth, if attained, would enhance the GDP growth rate and household income and expenditure, while reducing activity improves the Kuwaiti population ratio and reduces the rate of loss (per Kuwaiti) of public reserve funds held abroad. The author suggests a strategy combining aspects of both these approaches, with emphasis on micro-level diversification of the non-oil productive base.

Chapter 13, by Lance Taylor, uses a simple one-sector CGE elaborated with detail on financial and inflation accounting to explore options for price stabilization in Argentina. Among other complications it is assumed that (1) consumption demand responds negatively to the inflation tax (and thus rises when inflation is reduced); (2) markup inflation responds strongly to foreign exchange shortages, while wages are fully indexed; (3) there is cost-push on working capital finance due to higher interest rates; (4) exports tend to be crowded out by increases in domestic demand.

The model simulations suggest there is no need for extremely tight monetary and fiscal policy after a “heterodox shock” price freeze aimed at braking inflation; however, a surge in consumer demand from reducing the inflation tax requires attention. More fundamentally, markup inflation is likely to continue unless the country’s need to make a large external transfer to service debt is somehow reduced.

Notes

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1. Free translation by the author.

2. In model exercises that blend neoclassical and structuralist formulations, the trade policy simulations by deMelo and Robinson (1982) represent a partial exception.

3. The word “causality” is used here to describe the selection of variables that are exogenous (or “predetermined”) and endogenous in a set of equations or a model, combined with analysis of how the endogenous ones are influenced by the exogenous ones and themselves. Statistical notions of causality, according to which one variable “causes” another if it leads it in time, are ignored as irrelevant or misleading. Local earthquakes below the slopes do not “cause” a subsequent volcanic eruption, and money can either lead or lag output depending on the stance of the central bank, as Davidson and Weintraub (1973) point out. Pyatt (1988) discusses causality and closure issues from a perspective like the one adopted here in the context of a SAM.

4. Note 7 signals a case in which the relative magnitudes of the two savings rates matter. If wage earners (improbably) save at nearly the same rate as rentiers, then real wage cuts can be expansionary in a closure in which output is determined by aggregate demand; with a big differential in savings rates, overall demand can fall. Also nonzero saving from wages affects dynamic income flows. If they save, then presumably worker households have accumulated assets in the past, the returns from which should enter their income statements. We ignore this complication as being empirically unimportant, although it can in principle influence dynamic behavior of the macro system. In models with diverse asset/liability specifications, Pasinetti (196.2) and Taylor (1988c) show that assets held by wage earners do not substantially alter the nature of long-run macroeconomic equilibrium, so long as “significant” differences in saving propensities and/or portfolio compositions persist.

5. We omit taxation terms in the present discussion to keep the algebra as simple as possible

6. Note that in the short run the capital stock K is fixed so that it serves only as a scaling factor. Dividing nominal flows by the value of potential output or full capacity (e.g., $P\bar{K}$, where $\bar{K}$ is a technical parameter giving the maximum output that can be produced by a given capital stock) would be an alternative normalization. We don't adopt it simply to avoid carrying along the extra term K .

7. Note that this is a case where the relative magnitudes of saving rates from nonwage and wage income have a significant influence on the characteristics of macroeconomic equilibrium.

8. The markup is figured on costs exclusive of national intermediates. If $\tilde{\tau}$ is the rate on costs including such purchases, then

$$\tau = \frac{\tilde{\tau} + (1 + \tau)a_1}{1 + \tilde{\tau}a_1}$$

9. If service payments such as remittances and interest are carried in the accounting, the current account as opposed to the trade surplus is more naturally considered to be fixed by available capital inflows.

10. Chapters 10 and 11 in this book contain applications. See also Bourguignon, Branson, and de Melo (1988).

11. One obvious omission is disaggregation of the central and commercial banks in place of the “banking system” to permit analysis of monetary policy. Also households might usefully be disaggregated to reflect their different income flows and asset holdings, for example, workers, rentiers, and (with some practical interest) enterprise-owning households who can borrow from their own firms to engage in capital flight.

12. By increasing the government's demand for loans, public capital formation may crowd out private investment by bidding up interest rates in this specification. Recent evidence (Taylor 1988a) suggests that public investment in

fact crowds in private investment via complementarity effects. Tracing the financial implications in a CGE would be a useful exercise.

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